A Review of the Quality of Positioning Service of VLP Within the Indoor Positioning System Landscape

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***Abstract*—The rising demand for location-based services (LBSs) is accelerating the deployment of indoor positioning systems (IPSs) and driving the need for stricter quality of positioning service (QPoS) requirements. Leveraging existing illumination infrastructure, visible light positioning (VLP) is a promising low-cost candidate for centimeter- to decimetre-level localization, but its real-world QPoS—spanning accuracy, availability, latency, robustness, scalability, energy use, and cost—is not yet well characterized. Existing IPS literature, including VLP studies, mostly prioritizes accuracy, and no comprehensive, literature-based QPoS comparison of positioning technologies currently exists. This article presents a focused review of VLP from a broad QPoS perspective and situates it within the IPS landscape. We propose a reproducible, normalized QPoS framework to enable holistic, fair cross-study comparisons. Through a review of published experimental and simulation studies, we analyze how different classes of VLP approaches perform with respect to QPoS and position them relative to other IPS technologies, including ultrawideband (UWB), Wi-Fi/Bluetooth low energy (Wi-Fi/BLE), and vision/inertial measurement unit (vision/IMU) hybrids. It is shown that VLP can exceed the accuracy of many IPS technologies and can approach UWB-level performance, with a favorable overall QPoS under line-of-sight conditions and a preexisting suitable lighting infrastructure. Nevertheless, its suitability remains application-specific, as no IPS technology—including VLP—optimizes all QPoS dimensions simultaneously, and performance varies with deployment and calibration. Our review highlights several unresolved challenges in VLP research, including nonstandardized QPoS metrics, a lack of reference testbeds and protocols, limited 2.5-D/3-D and mobility evaluations under realistic multipath and industrial lighting, hardware and transmitter limitations, ambient-light interference, shadowing and multipath effects, and unresolved latency–reliability tradeoffs. To address these challenges, we propose future research on exploiting VLP signals of opportunity, optimizing transmitter planning, reducing driver/modulator cost, and characterizing hardware nonidealities, advancing multi-photodiode (PD) receiver studies, hybridizing with other systems, exploring reconfigurable intelligent surfaces (RISs), and developing standardized benchmark suites and protocols.**

***Index Terms*—Indoor positioning systems (IPSs), quality of positioning service (QPoS), visible light positioning (VLP).**

Received 14 November 2024; revised 2 December 2025; accepted 24 December 2025. Date of publication 20 January 2026; date of current version 5 February 2026. The Associate Editor coordinating the review process was Dr. Yuya Koyama. *(Sander Bastiaens and Morteza Alijani contributed equally to this work.) (Corresponding author: Morteza Alijani.)*

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Digital Object Identifier 10.1109/TIM.2026.3655932

I. Introduction

OVER the last decade, indoor positioning systems (IPSs) have become essential for locating people and assets in global navigation satellite system (GNSS)-denied environments, enabling real-time tracking, navigation, and context-aware services [1], [2], [3], [4], [5], [6]. IPSs are widely used in Industry 4.0 for asset tracking, workflow monitoring, and lean manufacturing in flexible factories and warehouses [2]; in smart buildings to optimize energy use and comfort via occupancy-aware heating, ventilation, and air conditioning (HVAC) and lighting control [3]; in Health 4.0 to manage patient flow, staff safety, and equipment tracking [4]; and in Agriculture 4.0 to guide greenhouse robots and monitor plants, workers, and tools [5]. Despite this potential, the indoor location market remains fragmented and underexploited, as legacy IPS deployments often fail to deliver sufficient QPoS when accuracy, robustness, coverage, latency, and maintainability are considered jointly. Comparative reviews consistently highlight strong tradeoffs between accuracy and system cost (hardware, cabling, installation effort, calibration, and maintenance) across technologies: for example, Bluetooth low energy (BLE) and Wi-Fi solutions are relatively inexpensive and easy to retrofit but typically offer only meter-level accuracy, whereas Ultrawideband (UWB) systems can achieve decimeter-level performance at the expense of higher infrastructure cost and tighter calibration requirements [7]. Moreover, there is a broad consensus that no single indoor positioning technology can simultaneously optimize cost, accuracy, robustness, and scalability, and that this has prevented the emergence of a de facto standard solution, leading to a market in which multiple partially incompatible systems coexist and are tailored to specific use cases [8].

Within this landscape, IPSs are commonly grouped by physical carrier, with radio frequency (RF)-based and optical-based systems forming two major families: RF-based approaches (e.g., radio frequency identification (RFID), wireless local area network (WLAN), ZigBee, BLE, and UWB) exploit received signal strength (RSS), time-of-flight (ToF), time-difference-of-arrival (TDoA), and/or angle-of-arrival (AoA) measurements, and have been extensively benchmarked in terms of accuracy, cost, and energy consumption [9]. Optical systems, in contrast, rely on visible or infrared (IR) light emitted by luminaires or dedicated transmitters. In particular, visible light positioning (VLP) uses modulated light emitting diode (LED)

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luminaires as positioning beacons, with photodiodes (PDs) or image sensors estimating position from RSS, time-of-arrival (ToA), AoA, or geometric constraints. This LED-based IPS paradigm can reuse existing lighting infrastructure and offers high spatial resolution, immunity to RF electromagnetic interference, and low incremental deployment cost. As such, it forms an attractive complement or alternative to RF-based systems in many indoor scenarios [10], [11], [12], receiving increasing research interest, as attested in [11]. Several surveys have reviewed VLP principles, algorithms, and performance (mainly accuracy) results, but there is still limited work that systematically compares VLP against other IPS technologies from a full QPoS perspective and positions VLP within the broader indoor positioning landscape [10], [11], [12], [13], [14], [15]. This review addresses that gap by evaluating the QPoS provided by VLP and situating it among competing IPS solutions.

The remainder of the article is structured as follows. Section II describes the state-of-the-art of VLP accuracy, as well as VLP's associated challenges and accuracy error sources. Section III describes QPoS metrics for VLP. Section IV discusses the current research areas and VLP's potential versus other IPSs. Section V discusses open problems and potential perspectives of VLP systems. Finally, conclusions are drawn in Section VI.

II. State-of-the-Art of VLP Accuracy

Inertial navigation system (INS), RFID, WLAN, ZigBee, BLE, UWB, and VLP are among the technologies competing for the indoor positioning market, each characterized by their own positioning error. This positioning error can be categorized into its two sources, namely, the precision and the accuracy. "Precision" refers to the proportion that can be attributed to noise, whilst "accuracy" denotes the systematic errors. The precision of a collected time series at a common measurement location is obtained via the Euclidean distance of each measurement point to the series centroid. Often though, the term "accuracy" is often used interchangeably with positioning error as a "pars pro toto," also in this work. In most works, it is still focused upon as the main QPoS metric.

VLP is claimed to be centimeter-level accurate, which would be considerably superior to most RF counterparts [16], [17]. The accuracy of an IPS can be affected by a variety of factors. System components such as transmitter and receiver, the localization configuration and environment, and localization algorithms can all have an impact. The most often used transmitter in VLP is the LED of which the emitted light power, radiation pattern and the deployment density strongly impact the IPS accuracy [11]. A PD or an imaging sensor (camera sensor) can be employed as the receiver device. The most common receiver component is the PD, but ambient light from the sun or from other illumination sources, multipath reflection, shot noise, thermal noise, PD tilt, or dust accumulation all can negatively influence the positioning accuracy. Notably, because it uses light as a propagation medium and relying on intensity measurements, VLP should suffer less than RF technologies from multipath reflections, further due to the stronger attenuation of reflected paths. Finally, the localization algorithms (e.g., multilateration, proximity, model-based fingerprinting (MBF), and multiangulation) also affect the VLP accuracy, and are to be considered in conjunction with the receiver configuration.

A. VLP Positioning Approaches

Primary localization principles in VLP systems [11], [13], [15] leverage RSS [18], [19], AoA [20], [21], [22], ToA [23], or TDoA [24], [25] measurements. ToA and TDoA are less commonly used because they require precise time synchronization [11], [13], [15]. AoA-based VLP systems use dedicated receiver configurations to estimate the angle of incidence from each of the LEDs, but at the cost of increased receiver complexity and cost [15]. Among all, RSS is the most widely adopted thanks to its low implementation cost and simplicity [11], [13], [15], usually using a single PD receiver to estimate the position relative to intensity-modulated (IM) LED infrastructure [11], [13], [15], [26]. Its principle relies on the fact that RSS decreases as the distance between the LED and PD increases.

Depending on the chosen localization principle, a wide range of algorithms has been employed [11], [13], [15], [26]. For example, multiangulation combines AoA information from multiple transmitters to estimate the receiver's location [20], [21], while multilateration relies on distance estimates derived from RSS [18], [19] or ToA measurements [23]. In addition, fingerprinting [24], [27], [28], [29], proximity-based algorithms [30], [31], [32], or other methods [33], are commonly deployed. Furthermore, data-driven approaches, such as supervised machine learning (ML) and deep learning [34], [35], [36], have been applied in VLP [13]. The choice of positioning principle and algorithm impacts the overall QPoS, depending on the specific application requirements and deployment conditions [11], [20]. Section II-B and II-C discusses both commercial and academic VLP deployments.

B. Commercial Deployments

The first noteworthy commercial deployments of VLC(P) are the trials of Koninklijke Philips Lighting (now Signify) with their Interact Indoor Navigation [37] solution in supermarkets, such as Carrefour in Lille [38]. Signify's system is also deployed in branches of EDEKA Parchman in Germany and Aswaaq in Dubai, United Arab Emirates. The solution exploits the AoA with a camera, in tandem with accelerometer and gyroscope data, in response to ceiling-mounted luminaires transmitting 16-bit identifier codes. However, not many details are found on its localization performance, as is the case for other commercial systems. In [39], the authors allude to a typical accuracy of 30cm with Apple iPhone 7+ positioning at a height of 1.2 m in a $100 \times 40 \times 7.7$ m environment, but no experimental results are given. Providing a seemingly similar solution, Lucibel claims to achieve a 10 cm accuracy [40]. ByteLight followed a similar approach to achieve a submeter accuracy aimed at the retail sector, and was acquired by Acuity Brands in 2015 [41], [42].

C. Academic Evaluations of Real-Life Deployments

Whereas Section IV-B will adduce the impracticality of benchmarking between IPS technologies based on the

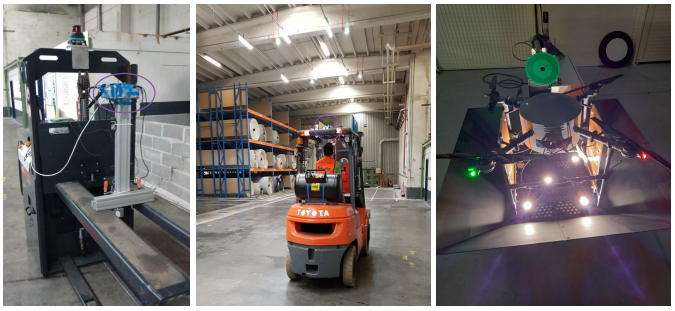
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literature, its arguments also apply to the comparison of the different VLP systems, especially, as most experimental data have been collected in small-scale setups. The goal of this section is to provide an overview of the important results where experiments are performed *on a relevant scale* and with a practical transmitter density. We mainly focus on experimental outcomes as simulation results typically do not port to real-life deployments [43]. Therefore literature is referred to [11], [14], [44], [45], and [46].

Analytical studies have shown VLP's bright prospective [47]. Its subdecimetre-order accuracy projection arises from calculating the Cramér–Rao bound, e.g., for RSS in [48]. Various research systems have already demonstrated VLP's viability [44], e.g., Epsilon [49], Luxapose [22], and Lightitude [50]. Their positioning accuracies are usually obtained in idealized environments, and with limited receiver-LED distances [26], [44]. The small studies, covering up to a $2 \times 2 \times 2$ m volume, display a centimeter-order ranging and (2-D) localization accuracy [39], [51]. The results of [52] clearly indicate the disparity between these results and practical deployments, despite assertions of the contrary [43]. For example, Alam et al. [52] reported a 1.9 and 16.1 cm median positioning error in, respectively, a 3.3×2.1 m laboratory and a 7.5×8 m open foyer environment, when employing spring-relaxation-based positioning.

The limited amount of experimental studies that consider a near-practical roll-out show at best a 10–35 cm positioning accuracy [14]. Li et al. [53] employ an RSS trilateration-based system in a $2.5 \times 2.84 \times 2.5$ m room to obtain a 90th percentile positioning accuracy (p_{90}) of about 25cm after sensor fusion (about 60cm for one-shot VLP). With a PD appended to a smartphone, a p_{90} of approximately 0.9, 1.1, and 1.8 m was reached in an open-plan office, a cubicle and a corridor environment, respectively [54]. Xu et al. [55] obtain a postprocessed mean error of, respectively, 6.0 and 8.7 cm in x- and y-direction, when localizing a commercial smartphone camera in 82 points underneath a high LED-density setup. Nakazawa et al. [56] equipped a 256×240 pixels camera with a fish-eye lens in order to achieve a reported minimal horizontal accuracy of 10cm in a $5.4 \times 7.5 \times 3$ m room. Finally, Raes et al. [57] investigated the use of ML for VLP based on RSS fingerprints in a 100+ m^2 industrial area with mm-level ground truth. Median positioning errors well below 10 cm are obtained, but at the expense of accurately collecting a substantial amount of training points, i.e., 50 or more per 24 m^2 LED quadrant.

Moreover, these results have mostly been obtained i) for 2-D (and thus not 3-D) positioning; ii) in the absence of both receiver [58] and transmitter tilt [59]; iii) in a conventional residential-type deployment; and iv) make use of prior propagation calibration measurements that will not be gathered in practice [60]. Furthermore, VLP is yet to be evaluated in the more stringent manufacturing or logistic environment, characterized by, non-Lambertian LEDs, higher ceilings, nonideal atmospheric conditions (e.g., dust), obstructions and metal reflectors [43]. All these challenges are pernicious for the VLP positioning performance. It can be concluded that, at this moment, VLP's positioning performance and practical communication range are not of the order that it is ready for many high-accuracy industrial-type applications. Finally, Table I compares representative academic and commercial VLP systems validated in real-world scenarios, highlighting their positioning accuracy, cost, deployment complexity, as well as key achievements and limitations.



(a) (b) (c)

Fig. 1. (a) LEDsTrack's world-first VLP-based route-configurable automated guided vehicle (AGV), (b) LEDsTrack solution for the monitoring of FLT driving behavior, and (c) InWareDrones' drone landing subsystem (video of the InWareDrones solution via https://www.imec-int.com/en/what-we-offer/research-portfolio/inwaredrones#video). The AGV LIDAR-based (proprietary) positioning system provides the ground truth for the VLP experiments.

D. Research Projects

1) LEDsTrack: The *LEDsTrack* research project[1] [65] aimed to realize a real-time, large-scale VLP-based location tracking solution, tailored to the industrial environment. The developed, cost-effective VLP-enabled tracking platform enables the roll-out of automatically reroutable AGVs, visualized in Fig. 1(a), and the analysis of forklift truck (FLT) driving behavior with collision or dangerous-driving alerts, as shown in Fig. 1(b). As such, the solution was aimed at the amelioration of the safety, flexibility, productivity, and cost-efficiency of industrial and logistic operations.

In tandem with accelerometer data, LEDsTrack's positioning results permitted to associate incidents (e.g., collisions), or either dangerous or noneco-friendly FLT driving behavior, e.g., in turning, to the FLT's location. The driver feedback was delivered using an FLT-dedicated touch screen, a smartphone application, or an external warning light or sound.

For AGV navigation with goods handling, a 10-cm accuracy was postulated as a viability bound, provided that additional highly accurate sensing systems are provisioned for pallet handling at the start and end of the route. Unfortunately, industrial VLP did not attain that 10-cm requirement.

Nonetheless, LEDsTrack showcased the practicality of the first AGV [see Fig. 1(a)] that drove with VLP as its primary localization system. Hereto, the chip-on-board (COB)-based VLP solution interfaced with the Dematic AGV's E'nsor[2] navigation platform, which employs, i.e., dead reckoning and odometry sensors. VLP served as an absolute positioning reference to help adjust the setpoints for the steering and

[1]https://www.imec-int.com/en/what-we-offer/research-portfolio/ledstrack

[2]Registered trademark.

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TABLE I
COMPARATIVE OVERVIEW OF REAL-LIFE ACADEMIC AND COMMERCIAL VLP SYSTEMS

System	Positioning Accuracy/ Metric/ Principle	System Cost	Deployment Difficulty	Main Achievements	Key Limitations
Philips [37]	∼30 cm Not available AoA	High (off-the-shelf phone + customized LEDs)	High (Requires precise alignment and denser luminaire placement (e.g., 1.5 m pitch [39]).)	Validated cm-level VLP accuracy in a commercial context	Works only with luminaires certified under the YellowDot program [39], [61]
Lucibel [40]	∼10 cm Not available Not available	High (off-the-shelf phone + customized LEDs)	High (Requires a VLC box between the LED and its power driver)	Enables real-time positioning using visible light in real-world environments	Requires direct LoS between camera and LED; Obstructions (e.g., pocket or bag) can block signal; Depends on user holding device facing upward
Acuity Brands [42]	$<$ 1 m Not available Not available	High (off-the-shelf phone + customized LEDs)	High (costly customized LEDs are needed for identity beacon implementation and replacement [62])	Real-time retail navigation and promotions	Requires direct LoS between camera and LED; Obstructions (e.g., pocket or bag) can block signal; Depends on user holding device facing upward
Epsilon [49]	∼0.4 – 1.1 m p_{90} RSS Trilateration	High LED modulation cost External light sensor board [63]	Moderate (Requires calibration)	Achieved p_{90} localization accuracies of: 0.4 m, 0.7 m, and 0.8 m in typical office settings; Good privacy preserving capability [63]	Performance drops when there are insufficient LED anchors. A single LED setup results in degraded accuracy (1.1 m) [49]; Requires external board connected to smarthphone [63]
Luxapose [22]	∼0.1 m p_{90} AoA	High (off-the-shelf phone + customized LEDs)	Moderate (Requires calibration and precise knowledge of LED positions)	Achieved high precision localization (decimeter-level) in both controlled and realistic scenarios	Requires a high density of LED deployment [63]; Too heavy computational overhead [63]
Lightitude [50]	∼1.93 m (office) ∼2.24 m (library) Mean RSS Multilateration	Low (no need for modulated LEDs)	Low (no need for modifying hardware or installing beacons)	Avoiding the need for expensive and labor-intensive site surveys; low cost and complex system	Received light strength (RLS) is highly sensitive to device orientation, which complicates accurate modeling and can impact reliability
Alam et al. [52]	∼1.9 cm (laboratory) ∼16.1 cm (open foyer) Median RSS Fingerprinting	High (luminaire with customized network connectivity)	Moderate (needs knowledge of LED positions and system calibration)	Provides a new alternative to traditional lateration and fingerprinting methods; Outperforms lateration, and competes favorably or outperforms fingerprinting depending on the environment	Simplified attenuation model; No motion tracking yet; does not account for device tilt
Li et al. [53]	∼25 cm (with fusion) ∼60 cm (single-shot) p_{90} RSS Trilateration	High LED modulation cost	Moderate (needs knowledge of LED positions and system calibration)	Fusion of VLC and Inertial Navigation and obtained cm-level accuracy	Initial position dependency; Potential orientation sensitivity
Xu et al. [55]	∼6.0 cm (x-direction) ∼8.7 cm (y-direction) Mean AoA	High LED modulation cost FPGA boards	High (needs knowledge of LED positions, power-hungry camera and image processing)	Obtained cm-level accuracy positioning using one smarthphone	The study is only valid for a fixed height with zero inclination; Needs further investigation under varying heights and inclinations
Nakazawa et al. [56]	∼10 cm (horizontal) Maximum Error AoA [64]	High (High-speed CMOS camera + fisheye lens)	High (Requires calibration; needs known IDs of luminaires and synchronization of VLC transmissions)	Achieved maximum horizontal error of 10 cm, validating high-accuracy positioning	Needs specialized receiver; Static evaluation only; No Mention of vertical accuracy
Raes et al. [57]	$<$10 cm p_{50} RSS Fingerprinting	High LED modulation cost	High (extensive fingerprinting and requires dense training)	Demonstrated robust performance of neural networks under challenging lighting and structural conditions	Accuracy relies on dense and stable RSS fingerprint maps; May require retraining or adaptation if lighting conditions change

driving control loops. A challenging trajectory was repeatedly, successfully traversed. This test showed that the developed VLP system, including the additional sensors, is suitable for sparsely deployed AGVs that mainly have a transportation function.

To assess the aptitude of industrial VLP, the 2-D positioning accuracy associated with the first industrial VLP system was evaluated in a 42 $\times$ 21 m storage rack-rich temporary depot zone at the premises of the Balta Group [66], a textile manufacturer. A p_{50} and p_{75} error of 29 and 58 cm, respectively, were observed with only the transmitter locations calibrated at ground level.

2) InWareDrones: The *imec.icon InWareDrones* project[3] targeted the realization of a completely autonomous drone-based solution for the inventory taking purpose [see Fig. 1(c)], significantly reducing the manual labor effort and the occurrence of stock errors when compared to traditional systems. VLP features in a newly developed precision drone landing system, a prerequisite for automatic and efficient wireless drone charging. As the inductive power transfer system's efficiency depends on its coil alignment, a precision landing system is required that is lightweight, accurate, and of low-cost. In [67], a landing system was proposed, in which a PD-equipped drone self-localizes with respect to LEDs that are integrated into the landing zone. For a 50 Hz update rate and a vertical range up to 1 m, a p_{75} error within 3.5 cm was attained with multilateration and a 6-element LED strip.

[3]https://www.imec-int.com/en/what-we-offer/research-portfolio/inwaredrones

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3) I2CAT: The *I2CAT* Foundation is a research and innovation center[4] that supports the technological transfer of research project innovations and intellectual property. They focused their VLP project on its applicability for disabled individuals, using the individual's location to provide information about his or her immediate surroundings. The system (LEDGO [68]) employs LED landmarks and a CMOS camera and is based on optical camera communication (OCC). It works well even when the receiver is not in direct line of sight with the light source [68]. Furthermore, it does not require any additional hardware and offers orientation in yaw, roll, and pitch.[5] A high positioning precision of 5 cm in 3-D mode and 3 cm in 2-D mode is claimed [68].

4) ELIoT: The enhance lighting for the Internet of Things (ELIoT) project[6] showed that Li-Fi systems provide both accurate positioning of devices and fast communication rates in environments with many devices. This is illustrated in an industrial scenario tracking the movement of an AGV utilized for the delivery of goods within a factory. It was found that Li-Fi-based location provides subdecimeter accuracy, with a mean error of 5 cm [69], and can be easily integrated into Li-Fi communication systems without extra installations.

5) Positioning Using Light Signals of Opportunity for Room-Reliable Asset Tracking (PULSAR): The imec.icon PULSAR project,[7] started in 2024, relies on uVLP and inertial measurement unit (IMU) fusion to determine and track a 1-m and room-reliable location of various assets in a hospital setting. The use of the existing lighting infrastructure is poised to significantly reduce the cost of deploying a large-scale IPS. Hereto, the characteristic frequency of existing LED luminaires is determined and used by a 2.5-D photoreceiver to estimate the tag's location [11], [70]. These location data are then to be leveraged to realize a generic solution for daily asset management and the optimization of care-logistic flows.

Some other initiatives have also investigated VLP, but rather as a side objective in a larger VLC-oriented research project. In the Internet of Radio-Light project (2017–2020) [71], the consortium targeted the realization of a safer, more secure, customizable and intelligent network that reliably delivers throughputs beyond 10 Gbps from access points pervasively located within buildings, whilst minimizing interference and electromagnetic field exposure and providing a positioning error of less than 10 cm in 95% of the performed simulation tests [72]. The European Training Network "European Training Network in Low-Energy Visible Light IoT Systems" [73] (ended December 2023) focused mostly on communication, modulation, and networking aspects, with only a few publications on VLP. An earlier Training Network "European Training Network on Visible Light-Based Interoperability and Networking" (ended February 2022), delivered a slightly larger VLP output, although its main focus was also on communication and channel modeling aspects [74].

[4]https://i2cat.net/
[5]https://i2cat.net/high-precision-3d-indoor-positioning-by-i2cat/
[6]https://www.eliot-h2020.eu/
[7]https://www.imec-int.com/en/research-portfolio/pulsar

E. Accuracy Error Sources

The above research projects and evaluations show that larger scale deployments suffer from errors larger than the centimeter-order that is often documented for simulations or experiments in idealized environments. With accuracy being regarded as the most important QPoS key performance indicator (KPI), we hereby discuss the main sources of accuracy errors. Table II lists a summary the VLP system's error sources and refers to relevant resources.

1) Transmitter Nonidealities: Reliability and practical concerns compel VLP to guarantee stable operation, even in the presence of transmitter nonidealities. These nonidealities comprise the LED's radiation pattern diverging from the tabulated one [84], LED aging [78], nonideal LED modulation [80], and fluctuations in $P_{t,i}$ [77] that are inherent to the either lamp or the overarching electricity grid's loading. In [75], it is reported that magnitude swings of up to 38% of the average level are possible in an LED's radiant flux $P_{t,i}$. Using photocurrent $I_{\mathrm{PD},i}$ redundancy positioning algorithms, $P_{t,i}$ variations of a few LEDs can be accommodated. Though in the process of installing the lighting, whether VLP-enabled or not, it is assumed that common practices are wielded to avoid any visible tilt. Unfortunately, even small, unnoticeable tilts impact VLP, especially at the industrial scale [59]. There, for example, a one-degree off-axis tilt induces a maximum power point ($I_{\mathrm{PD},i}$) shift of more than a decimetre at a 6-m LED-PD-plane-distance. Actually, as shown in [59], the extent that tilt influences the positioning also depends on the directivity of LED sources and other environmental parameters.

2) Receiver Noise Floor and Orientation: Evidently, the positioning performance is determined by the receiver parameters, such as the noise-equivalent power. Receiver tilt has a detrimental impact too [58], [96], [97], [98]. Luckily, in 2-D localization, with small ($<2^\circ$) random tilts or vibrations, receiver tilt is not the accuracy-limiting factor. This statement finds its roots in a trajectory simulation in the virtualised environment of Section II-D1. Employing a camera or multiple PD-receiver [86] and/or IMU-based compensation techniques [58] permits mitigating tilt's influence.

3) Ranging Bias and Calibration Fitting: Range/location bias manifests for RSS VLP, in the form of nonideal (Lambertian) acceptance [60] or radiation patterns [84], of tilting and of time-variations on the radiant power, whether present purposefully with dimming or not [11], [77]. The bias factors are frequently (arbitrary) calibration fit at once [99] or indirectly learned with ML models based on tedious in situ measurements [94]. However, these manual measurements are superfluous when properly modeling [60]. Range/location errors are also incurred due to LoS blockage [100], attenuation/refraction with partially opaque objects, and reflections-induced non-line-of-sight (NLoS) that is moreover not easily mitigated being devoid of an accurately determined channel impulse response (CIR). Delivering the required high accuracy necessitates the measuring and fitting of the distance d_i - RSS (i.e., $I_{\mathrm{PD},i}$) relation. Such an approach is definitely not feasible for large-scale VLP roll-outs. This is both due to the laborious nature of the 'fingerprint' gathering, as well as

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TABLE II

LISTING OF THE VLP SYSTEM'S ERROR SOURCES, ALONG WITH REFERENCES TO IMPACT STUDIES AND AN OVERVIEW OF MITIGATIONS ($P_{t,i}$ = RADIANT FLUX OF LED$_i$, $I_{\text{PD},i}$ = PHOTOCURRENT GENERATED DUE TO LED$_i$, M = LED-PD MISMATCH GAIN, $R_P(\psi_i, \omega_i)$ = RECEIVER RESPONSIVITY, $s_i(t)$ = TRANSMITTED SIGNAL, I_{LED} = LED DRIVING CURRENT, $I_{\max}$ = MAXIMUM LUMINOUS FLUX, AND RRSS = RATIO RSS)

	Source	Type of induced error	Treated in []	Methods for Mitigation
Transmitter	■ Calibration error on $P_{t,i}$ and/or transmitter location coordinates ■ DC/AC instability of I_{LED} due to LED driver or grid emission ■ Device-to-device variation ■ LED-aging ■ LED outage or $P_{t,i}$ reduction ■ LED replacement ■ Nonideal modulation – Overshoot and non-zero rise/fall transitions – Nonorthogonal $s_i(t)$ in dense deployments or for limiting modulation loss – Clock drift and jitter ■ Non-point source LED and/or non-Lambertian radiation pattern ■ Radiation pattern divergences – e.g., Lambertian order $m_i \neq 1$ ■ Uncompensated transmitter tilt	■ Bias ■ Variation on $P_{t,i}$ up to 38% [75] Noise source ■ Device-dependent, nonnominal $P_{t,i}$ needs calibrating or leads to bias ■ Large time-scale diminishing $P_{t,i}$ and M impacted by colour-shift, bias ■ Large time-scale $P_{t,i}$ reduction, bias ■ Potential bias with changing $P_{t,i}$, and even radiation pattern and M ■ Nonnominal $P_{t,i}$, $s_i(t)$ or $I_{PD,i}$ leakage to $I_{PD,k}$ $k \neq i$, i.e. inter-LED-interference. Bias and noise source. ■ Non-monotone $d_i - I_{PD,i}$, different positioning gradient and SNR distribution ■ Bias ■ Bias	■ [76] ■ [77] ■ [77] ■ [77], [78] ■ [77], [78] ■ [76] ■ [79], [80] ■ [81]–[83] ■ [84] ■ [85], [86], [59]	■ Not easily mitigable for (R)RSS with 1 PD. For $P_{t,i}$, AOA helps. ■ Not easily mitigable for (R)RSS with 1 PD. AOA does. Infrastructure PD with known position to keep track of fluctuations. ■ The variation degree may be tolerable, otherwise the I_{LED} - I_{max} relation needs calibrating. ■ Using an ageing process model or secondary PD with fixed position, in combination with scaling the $P_{t,i}$ in the positioning. Can be autocalibrated when a lamp-integrated PD measures the floor reflection. ■ Detectable, e.g., on the basis of the location variance across various LED subsets, or by computing $P_{t,i}$ estimates from a location tracked positioning estimate. When detected, discard LED$_i$, e.g., by employing a binary state model over time. ■ Profound changes most likely require a recalibration. ■ Important issue to tackle: – Improve LED driver design or calibrate – Recursive demodulation or incorporate in model-based fingerprinting propagation model – Improved $s_i(t)$ assignation such as [87]. – Improve modulator, e.g., use same, stable crystal oscillator as in receiver. ■ Accurately calibrate and model transmitter e.g., with ray-tracing based or composition propagation models. ■ Accurate characterisation or ranging correction model. ■ Recommendation to hang LED as precise as possible. If possible accurately calibrate in situ and integrate in propagation model.
Channel	■ Ambient light including infrared light on PD ■ NLOS influence – Multipath – LOS blockage and/or refraction – Transmission transparent obstacles ■ Polluted atmospheric conditions, e.g., fog or airborne dust	■ Dominant noise contribution, potential interference with $I_{PD,i}$ ■ Bias error due to changing $d_i - I_{PD,i}$. Multipath effectuates $I_{PD,i}$ surplus. Outlying reduced $I_{PD,i}$ due to blockage. Obstacle transmission reduces $I_{PD,i}$. ■ Changing d_i - $I_{PD,i}$ due to scattering, according to Beer-Lambert law.	■ [88] ■ [89], [90], [91], [43], [92] ■ [43]	■ Noise averaging and visible light spectral filter, adaptive gain control or DC cancellation circuit [93] to cope with large DC swing. With PD saturation: a narrowband interference filter or tilter/rotator. Optionally identify the source with of colour and/or IR sensing, for cancellation. ■ If possible account for in network planning and/or propagation model – Location filtering with sensor fusion. – Use NLOS estimation, discard/selection or mitigation method – Rule-based blockage detection to discard $I_{PD,i}$ and/or location filtering – Location filtering ■ Estimate attenuation with fixed detector. Though, best utilise RRSS (or AOA). Employ statistical and/or dedicated propagation model.
Receiver	■ Nonideal Signal acquisition – Nonnominal sampling frequency – Clock drift and jitter – Hardware limiting effective number of bits ■ Device-to-device variation ■ Dust or other obstructions ■ High velocity mobility and Doppler effect ■ Improper $R_P(\psi_i, \omega_i)$ model ■ Short-time pose variation ■ Tilt offset ■ Unknown dimming	■ Non-coherent detection of $s_i(t)$. Inter-LED-interference due to $I_{PD,i}$ leakage. Bias and direct noise source. ■ Device-dependent, nonnominal M or receiver responsivity needs calibrating ■ Attenuation of all $\{I_{PD,i}\}$, bias ■ Bias due to nonnominal M ■ Bias ■ Noise source ■ Bias ■ Large time-scale variation on $I_{PD,i}$	■ [79], [80] ■ [91] ■ [43], [94] ■ [95] ■ [60] ■ [85] ■ [85], [86], [60] ■ [78]	■ Optimise hardware design, especially the clock module for ADC and (de)modulation. Account for expected inter-LED-interference in software. ■ Characterise. Depending on extent, resort to RRSS (or AOA). ■ Employ RRSS (or AOA) ■ Ensure a high update rate. Doppler influence is limited indoors and typically not mitigated. ■ Good characterisation of $R_P(\psi_i, \omega_i)$. Then, depending on i.a. the required update rate and expected propagation angles, identify the best approach of responsivity model. ■ Robust localisation, especially location filtering ■ Employ gimbal-type compensation to avoid tilt. Large tilt compensation based on SPAO ([96]) or with IMU information, e.g. with [58] or sparse model recomputation [78]. Again, SPAO-type algorithms benefit from angular diversity (e.g., AOA [86]). ■ Communication of dimming information. Employ fixed position reference detector. Estimation from waveform if possible, e.g., based on guard spaces or duty cycle estimation with pulse-width modulation.

the impracticality in foreseeing a temporary accurate-enough ground truth system for that.

4) Ambient Light: Natural sunlight, non-VLP lamps' artificial light, and even the VLP-enabled transmitters' biasing contributions and spurious modulation by-products all fall under the ambient category. Besides a dc component, the electrical frequency spectrum of a lamp, even when purely meant for illumination, has a strong representation of the power line frequencies and their harmonics. Hence, ambient light's spectral content is typically limited in frequency. It is, however,

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characterized by a dc component of a large magnitude. The disadvantages associated with ambient light are that it adds to the receiver's (shot) noise, that it may exert interference, and that it has the potential of saturating the receiver to effectively "drown" the VLP signals. The consequence is a degraded positioning performance. Particularly, ambient sunlight poses a viable threat to VLP systems [101], as its luminance is typically 2 orders of magnitude larger than that of the typical light source. Longer VLP links, which are associated with more severely attenuated signals, are comparatively more vulnerable to ambient light, than is the case in residential-type environments. To better cope with ambient light, several solutions can be thought of: i) enlarging the directionality of the VLP link; ii) extensive channel averaging (e.g., using phase shift detectors) or coding [102]; iii) electronic circuit-level compensation [103], [104], such as adaptive gain control or with a dc current cancellation and low-frequency filtering circuit [93]; iv) employing an optical (attenuator) filter; or v) using a micro-electro mechanical system (MEMS)-based tilting platform to move the sensor away from the ambient light source.

F. Particular Challenges in Industrial Environments

Despite the above error sources, VLP seems to be equipped to differentiate from its IPS competitors in industrial environments, by supporting a high scalability, a high accuracy, and a high update rate, while limiting the cost. Moreover, the nature of the environment and its associated activities is such that typically IPS cannot assume the presence of a smartphone for localization. As a result, employing a dedicated PD-receiver does not present an economic hurdle. The combination of these factors renders PD-based VLP well-suited for both vehicle or (moveable) asset tracking and ground-based (autonomous) indoor vehicle navigation. Both application categories come with vast economical and well-being incentives (see Section IV-A). The tracing and efficacious guiding of the ever-growing fleet of AGVs, for example, enables more effectively automated infrastructures that then offers benefits in the form of vast energy savings, an enlarged productivity, additional safety and a minimization of the operating costs. The only allusion regarding industrial VLP's performance is done by Kouhini et al. [39] on the earlier mentioned signify VLP solution (see Section II-B). However, the lack of transparency, regarding, i.e., the required calibration, the update rate, the system cost, and the one-shot VLP accuracy, impedes from building on those results. The system also requires a smartphone, which comes at its disadvantages.

Unfortunately, the dynamically varying industrial or warehouse setting does not present an easy environment for (visible light) positioning [43]. It requires systems that remain fully functioning, even in the presence of positioning-disruptive specific factors, such as the noise bursts that arise due to a changing dust concentration and/or blockage [105]. The QPoS requirements tend to be more strict as well, compared to those of consumer LBS. Based on the study of Berenguer et al. [105] on the signal-to-noise ratio (SNR) experience during 8×6 MIMO channel measurements in a manufacturing cell, it can be concluded that applying VLC and thus VLP in industry is in fact feasible. Nevertheless, the industrial-type of environment sets forth several high-accuracy hindering factors for VLP [82]. To bridge the gap between the small-scale lab testing and the industrial domain, industrial VLP needs to sustain highly accurate positioning among the following challenging environment factors.

1) Larger Inter-LED-Receiver Distances: The industrial-type environment typically has higher ceilings. The associated larger link distance entails a lower SNR, and in turn a decreased positioning performance [16], [48], [106], [107], [108]. Without tilt, the distance-dependent ranging bias is limited [85].

2) Non-Lambertian Radiation Patterns: In the industrial or warehouse hall, illumination is generally not provided by chip on board (COB) or standalone surface-mounted device (SMD) LEDs, which can be approximated as ideal Lambertian radiating point sources and allow a relatively simple localization [14]. In industry, linear LED fixtures, luminaires with nonpoint-source-like dimensions are used instead to meet the stringently imposed and well-regulated illumination bounds, e.g., in terms of glare or minimum illuminance [109], [110], [111]. The LED fixtures generally combine 1 or more (SMD) LED strips with diffusers to transmit more radiant power more uniformly, while also allowing for glare control. These devices can no longer be modeled as point sources [11], an assumption frequently required for localization, especially for multilateration.

3) Nonideal Atmospheric Conditions: Depending on the economic activity, the atmospheric conditions in the industry may prove challenging. The higher dust or fog concentrations [43] also diminish the signal detection range. Raes et al. [94] characterized the influence of dust accumulation on the PD. Without dust, an optimized Gaussian process regression-based localization yielded a 2-cm median positioning error. In the presence of chalk and saw dust, this error degraded to 10 and 14 cm, respectively. For multilateration, the performance loss was even more pronounced (median error exceeding 30 cm).[8]

4) Shadowing and Multipath Effects: The industrial/logistics environment typically heavily features positioning obstacles, ranging from human workers, over machinery to storage racks [43], most often metallic in nature. These introduce both shadowing and multipath effects in the visible light channel [112], (locally) impacting positioning performance. Bastiaens et al. [113] demonstrated that NLoS influence occurs mostly locally around obstacles and specifically around the specular reflection vector of each LED. Current research has not yet thoroughly assessed the impact of multipath on VLP accuracy, especially due to the dynamically changing setting [43]. According to [114], [115], [116], and [117], multipath reflection reduces indoor positioning accuracy, and edge and corner regions have higher positioning errors than other areas. In case of (temporary) light signal blockages, fusing VLP data with IMU data using dead reckoning techniques could serve as a mitigation.

5) Feasibility of Scalable Localization: Accounting for some of the previous, noninvertible propagation factors does

[8]Schneider et al. [112] also report on a case study on the to be expected dust and the optical crosstalk influence.

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not befit most of the localization algorithms. As a consequence of these factors, the relation between the LED-to-PD distances $\{d_i\}$ and the RSS values $\{I_{\mathrm{PD},i}\}$ becomes ambiguous. Therefore, MBF localization can be turned to [11], but requires storage reduction methods to reduce positioning estimation latency. Moreover, localization methods for industrial applications are required to ensure adequate robustness and availability as well. This is where composition, location tracking, and sensor fusion approaches come in.

III. Quality of Positioning Service (QPoS)

As said, accuracy is still the main considered metric, here defined in terms of the 2-D or 3-D Euclidean distance between the estimated $\hat{\mathbf{x}}$ and actual ground truth location $\mathbf{x}$. To adhere to prescribed good practices [118], the 50th/75th/90th/95th percentile $p_{50}/p_{75}/p_{90}/p_{95}$ of the positioning error are employed as summarizing metrics. When limiting the analysis, frequently, the p_{50}/p_{90} combination suffices. The p_{95} is typically already heavily influenced by the measurement [118] and is mostly relevant in conjunction with a highly accurate ground truth. For a single metric, the p_{75} intrinsically contains the most information [118].

However, to ensure an in-depth analysis and benchmarking between different technologies, a more broadly defined QPoS [119] needs to be employed. Other QPoS KPIs besides accuracy are also relevant, but are currently only sporadically considered: the IPS' price tag that also incorporates the installation and maintenance effort, the safety and security, the coverage area and typical link distance, the interoperability, the degree of standardized or proprietary parts, robustness (consistency and tolerating perturbations), latency and availability [120].

Before providing a more general overview of how VLP is positioned in the IPS landscape with respect to different QPoS and other aspects, we first focus on a more in-depth comparison of (u)VLP and for UWB, two technologies with comparable accuracies [91] and (therefore) often targeting similar use cases. We will discuss the technologies' scalability, their update rate, robustness and availability, power consumption, required calibration and installation, price tag, the privacy, safety, and security aspects, and the standardization efforts and the commercial interest.

A. Scalability

As it impacts the update rate, two-way ranging (TWR)-based UWB anchor products come with a 5–10 tags recommendation. Therefore, TDoA-based UWB is inherently more scalable than the TWR variant [121]. Being broadcasting in nature, a standalone VLP has theoretically infinite receiver-side scalability. For Li-Fi systems, this is evidently not the case. Both technologies' transmitter scalability is enabled by frequency reuse, though limited by intercarrier interference/intersymbol interference. VLP's smaller propagation cells support denser networks. All in all, it scales better than UWB.

B. Update Rate

In RSS-based VLP, the update rate depends on the minimal modulation frequency spread, which in turn is influenced by the transmitter bandwidth and the frequency reuse. Update rates exceeding 100Hz are (easily) supported. Under accuracy constraints, the time-variability of uVLP [10], [11] prohibits these high-speed applications. The update of TWR-based UWB is contingent on the number of communication rounds, and the device count. Triggering a location update upon each new range measurement, 50Hz is attainable. TDoA UWB is then better suited for update rates >100 Hz [122].

C. Robustness and Availability

VLP still faces significant challenges regarding its robustness and availability. Tilt, LED outage, LoS obstruction, or NLoS-induced outliers quickly come to mind. Enabling the VLP service is sometimes also impeded, e.g., with single-row lighting deployments. In addition, the prolonged use of RSS-based (u)VLP entails dealing with the adverse effects of LED aging (and/or changeover) and dust accumulation on the PD (AoA-based VLP should be more robust). For uVLP [10], [11], the LEDs' characteristic frequency and magnitude variation [70] further reduces the robustness. On the other hand, UWB does have to deal with its issues, such as NLoS and radiation pattern-alteration by close metallic materials. However, its CIR allows devising more complex compensation measures. Therefore, it is clear that UWB is the more industrial deployment-ready of the two. Finally, regarding outdoor adaptability, whereas RF-based systems often function well outside (except for the possible impact of rain, for example), VLP is less suitable for outdoor usage as LEDs are typically ceiling-mounted, and the PD can suffer greatly from shot noise due to sunlight.

D. Power Consumption

System-wise, UWB is easily identified as the most power-conservative. The transmission and receiver operation can function with <10 and <20 mW, respectively. A 50–80 mA-consuming operation is common [123]. UWB's overall power consumption depends on its ranging principle. TWR typically requires multiple communication rounds, while TDoA only necessitates one. As in TDoA, the tag's range is computed by the anchors, some use cases require the range to be communicated back to the tag, requiring two communications.

Illumination-compatible lamp transmitters typically consume multiple Watts, impeding battery operation. Being always "on," even when maximally dimmed in the darklight configuration [124], [125], VLP is perceived to conflict with the trend of energy savings, especially in daytime in indoor environments benefiting from natural outdoor light. VLP's receiver power consumption is device-dependent. A solar panel receiver is energy-neutral. A typical PD receiver consumes approximately 50 mW. It should be noted that the PD is not the dominant consumption source. That "honor" is reserved for a combination of the electronic components that constitute the analog and digital receiver chain.

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E. Required Calibration and Installing

Depending on the environment, autocalibration vastly reduces the manual calibration effort required [126]. Interestingly, comparable autocalibration techniques have been proposed for light-based positioning as well [127]. However, the one-way nature of broadcasting VLP hampers their accuracy. In the considered VLP system, a lamp-integrated PD can serve to dynamically chart $P_{t,i}$ via reflections on the ground, provided that the latter's reflection properties are known. Overall, UWB is more install-friendly. It involves fewer anchors, no retrofitting, and no mains voltage interactions. UWB also better tolerates installation nonidealities, such as transmitter tilts.

F. Cost

Quantifying the systems' roll-out cost is onerous, as the minimally required infrastructure depends on the targeted QPoS. For example, whereas an average office can be adequately covered with four UWB anchors, warehouses with their metallic storage racks would typically demand a multitude. Furthermore, an IPS technology can be associated with multiple feasible solutions, e.g., with PD or camera receivers in VLP. Therefore, in this work, only subjective, ballpark cost figures are provided. However, the numbers are loosely based on the proprietary solutions of the companies involved in the LEDsTrack project (see Section II-D1). The area of study is the industrial environment shown in Section II-D1. light detection and ranging (LIDAR), Wi-Fi, UWB, and PD-based VLP are considered as IPS.

Only the physical systems and the installation/calibration cost are considered. Charges associated with other person-hours (e.g., in planning), the procuring of (micro) services (e.g., asset management), or operating expenditures are not accounted for. Moreover, an abstraction is made of the economics of scale.

1) Lidar: The cost of the LIDAR sensor, of the eight passive beacons and of the installation cost amounts to €3000, €80, and €600, respectively.

2) Wi-Fi: The Wi-Fi infrastructure provisions six anchors, each with a €450 and €50 component and installation cost, respectively. A €50 receiver (chip and external antennas) is employed.

3) Ultrawideband: 12-UWB anchors of €200 provide an adequate positioning coverage. An installation cost of €50 is charged for each anchor. Two receiver scenarios are considered, each with a different receiver unit cost: "−1" of €200 and "−2" of €50. The former rounds the component cost of €178.22 [123] up (e.g., for a casing cost), the latter corresponds to either a wearable and/or a batch-scaled variant of €178.22.

4) Visible Light Positioning (VLP): For VLP, 4 scenarios are discerned. Common to all is that 21 VLP-enabled LEDs are employed in Section II-D1 and that the receiver's price is €75. In Fig. 1, "−1" represents the case of an infrastructure being able to be used as-is. This includes both a compatible dimming interface being present or uVLP being utilized. Scenario "−2" refers to when an LED driver needs retrofitting or when

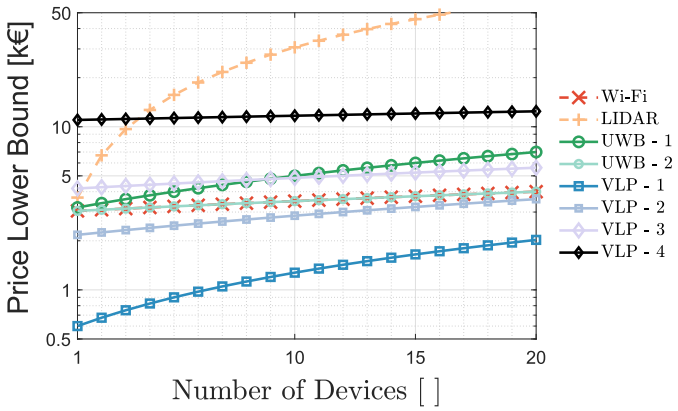


Fig. 2. Fictive cost projection study of LIDAR, Wi-Fi, UWB, and PD-based VLP in the environment of Section II-D1.

dedicated COB-type LEDs are placed, whilst "−3" denotes the case when six supplemental illuminance lamps are required. As the long lifetime of LEDs does not facilitate a quick overall infrastructure upgrade, a VLP-enabling fee will apply for both. Finally, "−4" represents the worst-case scenario of no prior illumination infrastructure being present. Then, 21 VLP-enabled and six ordinary lamps need to be installed. In "−1," a €25 amount per LED is levied for measuring the LEDs' frequencies. For "−2"/"−3," the charge for VLP-enabling is €50 and the new driver's component cost is also €50. An illumination lamp, not supporting VLP operation, fetches €300. A €35 amount is levied for their installation (no location calibration). In "−4," 21 VLP-enabled lamps of €350 take €75 to install.

Fig. 2 confirms the assertion that VLP's economical value strongly depends on the infrastructure. When either the existing lighting infrastructure can be used as-is or can be retrofitted/appended, VLP tends to hold an economic advantage over its competitors. The advantage will become more pronounced, when commercial endeavors will drive down the receiver cost. With illumination compensation lamps, the upfront investment rises. The economic viability will now depend on UWB/WiFi and even LIDAR. With the other QPoS metrics in mind, presumably, VLP will lose out to UWB. When VLP is not able to fall back on prior lighting, it is not the best option (unless integrated with Li-Fi). Fig. 2 highlights that LIDAR does not receiver-scale well, and mainly bodes well for a small fleet covering a large-scale environment.

Both Wi-Fi and LIDAR are included in Fig. 2 to demonstrate that the application potential of (u)VLP and UWB is delineated by its competitors. Wi-Fi/LIDAR is the dominant technology in certain market segments.

G. Privacy, Safety, and Security

The non-RF-based nature of VLP ensures a wide applicability, and admission to hospitals, mines, or airplanes. Lights are furthermore ubiquitous. UWB has been cleared for these types of environments as well. Standalone VLP is one-sided, meaning that only the receiver has knowledge about its position estimates. Privacy is found in that nor the infrastructure nor external observers are a priori aware. The spatial confinedness of light propagation effectuates physical layer security,

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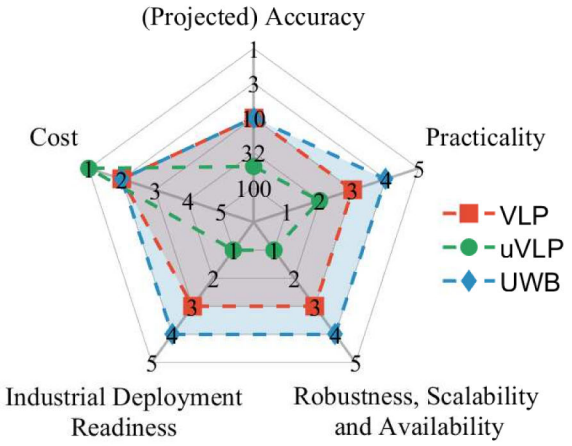


Fig. 3. Spider plot representation of the author's assessment of the current UWB and (u)VLP systems.

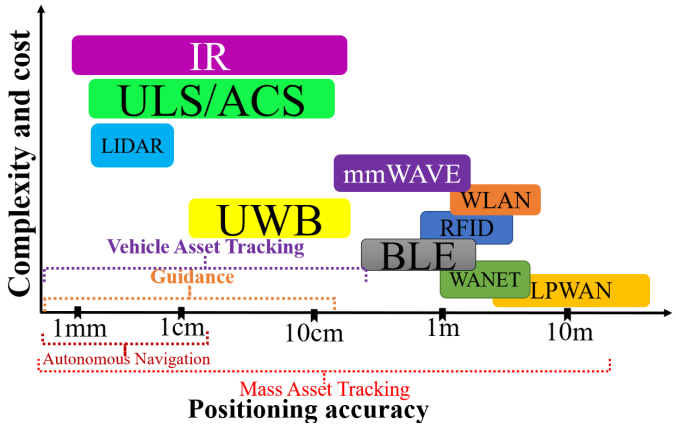


Fig. 4. Landscape of IPSs. Though in a truthful analysis, all QPoS dimensions need to be considered, this figure zoomed-in view on the two most important metrics, cost versus accuracy. The positioning latency dimension is represented by the thickness of the black edge of the rectangulars. This graph, differing from that of [152], denotes both the development speed of IPS and the subjective extrapolation nature of the IPS comparison.

requiring eavesdroppers to be in view. In its simplest form, VLP employs no encryption, though encryption is certainly supported, especially in VLC/Li-Fi [128]. Rogue transmitters and receiver saturation/jamming are effective threats.

Cryptography is well-engrained for UWB, as are other physical and MAC layer protections. UWB is resilient against jamming and exhibits a low probability of detection/interception. All are part of the rationale behind UWB being used for passive, keyless entry applications. Implicit or explicit impulse radar sensing can be in dissension with a user's privacy. For example, it can be used for the malicious, remote detection of presence and even vital signs [129].

1) Standardization and Commercial Interest: The interoperability, arising from the standardization efforts and the product-availability, that drives UWB's many sprouting (subsidiary) scale-ups, is not yet in place for (u)VLP. In contrast to multiply amended IEEE 802.15.4 (a/z-2020), VLC/Li-Fi standardization efforts have only recently resulted in the 802.11bb standard, focusing on physical layer specifications and system architectures for wireless communication using light waves, and interoperability with Wi-Fi [130]. Furthermore, with commercial deployments solely encompassing pilot studies (see Section II-B) and most hardware being prototypes, VLP is still firmly in the research stage.

Based on these QPoS pillars, whose average subjective scores are visualized in Fig. 3, the main takeaway is that UWB is much further along on its way to become a mature indoor positioning technology than VLP is. Fig. 4 demonstrates the landscape of available IPSs, in terms of their cost versus their accuracy.

Choosing the best VLP algorithm for a given application scenario depends on the roll-out requirements and the desired QPoS. Algorithms vary in terms of accuracy, power consumption, cost, and accuracy. Table III maps the most prominent VLP subtypes (proximity-based, fingerprinting, multilateration, and multiangulation) to their corresponding QPoS profiles, including latency (the delay between a receiver being in a certain position and the corresponding positioning estimate becoming available) [131] and complexity of design, implementation, and deployment of the VLP system,

TABLE III

COMPARISON OF VLP APPROACHES BASED ON QPOS ASPECTS

QPoS Aspect	RSS Proximity [30]–[32]	RSS Fingerprinting [27]–[30]	RSS Multilateration [94], [136]–[138]	AoA Multiangulation [21], [139], [140]
Cost	Low	Medium – High	Low	Medium – High
Latency	Low	High	Low	Medium
Complexity	Low	Medium	Low	High
Power Consumption	Low	Medium – High	Low	Medium – High
Accuracy	Low	Medium – High	Medium – High	High
Safety	Medium	Medium	High	High

encompassing both software and hardware aspects, as well as the computational resources needed for positioning [14], [120], [132], [133].

Proximity-based localization is a simple and low-cost VLP algorithm that provides a fast but coarse positioning within the coverage area of a single LED (typically a few meters) [139], [140], [141], making it suitable for real-time applications such as asset tracking in hospitals [26]. Compared to the other three approaches, it has a lower accuracy, but is able to relatively well maintain this accuracy under moderate perturbations (e.g., external noise and moderate PD tilt). The other methods provide higher accuracy, which becomes quickly compromised under nonideal conditions [142]. Fingerprinting-based positioning in principle has the highest accuracy [143], thanks to an extensive offline phase for collecting location data, followed by an online phase matching incoming measurements to the dataset using techniques like probabilistic models [29], k-nearest neighbors (kNN) [28], [144], or correlation [145], [146]. It can be effective in environments where multiangulation and multilateration methods are impacted by strong reflections, but it is usually not feasible in practice, due to the costly data collection phase, and the unfavorably high storage and latency characteristics [139]. While AoA-based methods in theory attain higher accuracies than multilateration-based approaches, they come at the expense of a more complex receiver structure.

H. Formalizing QPoS

QPoS aims to offer a holistic evaluation of a positioning system by integrating multiple indicators beyond accuracy.

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TABLE IV

CLASSIFICATION OF LBS BASED ON THEIR ASSOCIATED QPOS AND IPS (SI: SMART INDUSTRY OR "INDUSTRY 4.0," SA: SMART ARCHITECTURES, RE: RETAIL 4.0, HE: HEALTH 4.0, AG: AGRICULTURE 4.0, 6G: SIXTH-GENERATION TECHNOLOGY, SP: SPORTS, RPS: REMOTE POSITIONING SYSTEM, WHERE INFRASTRUCTURE CALCULATES POSITION, SPS: SELF-POSITIONING SYSTEM, WHERE TAG CALCULATES POSITION

Type	Incentives (sector)	Important QPoS KPIs	Current IPS	Status and Developments	Perspectives for VLP
Tracking Large Quantities of Small Assets	■ Reducing labour effort and increasing accuracy in asset taking [SI]; ■ Optimisation supply chain [SI,RE]	■ Large quantities; ■ Small form factor; ■ Near-negligible expense per unit; ■ Metre-order accuracy; ■ (Sub) second-order latency	RPS; Visual markers, e.g., barcodes & RFID tags	Well-provided for emergence long-lifetime and/or backscattering UWB and BLE tags to improve coverage, interference and ranging resolution [162]–[165]	■ Difficult to prevail against entrenched RFID systems; ■ Backscattering principle extends to VL(C)P, a voltage-addressable mirroring grating, or liquid crystal display-based switching [166], but difficult to envision their economic viability with silicon integration and/or additive manufacturing scaling not on the horizon; ■ Tagless RPS forms, VLS, face resolution granularity questions
Location service for Information Retrieval and Geofencing	■ Security and safety in media access control [SA,SI]; ■ Efficiency and targeting in location-based advertising; ■ Enhanced experience in museum guidance [SA]; ■ Non-internet-connected sensor networks (SN) e.g., to optimise cultivation processes, i.a. by health sensing of plants [167] or livestock [168] [AG]; ■ Supply chain optimisation by IOT [AG]	■ Inexpensiveness ■ Light-weightiness ■ Easy deployment ■ Accuracy and latency of secondary importance	SPS, RPS; QR codes or BLE beacons, with Wi-Fi or cellular off-loading; Occasionally WLAN-base; WANET for sensor networks	Well-provided for in light of its networking capabilities and smartphone integration; Cheap, low power and more accurate; future UWB/BLE beacons dispense with need for additional (WLAN) connectivity and more extensive market penetration is easily imagined	■ Current VLP systems are of littler avail, unless an overarching VLC/Li-Fi network is already installed; ■ best suit smartphone-based (u)VLP, in the form of OCC or with the integrated PD; ■ (u)VLP's confined coverage area would enable fine-grained spatial access control; ■ Without enlarging the receiver complexity, a battery-only VLP transmitter, a prerequisite to compete with the competitors, is infeasible.
2D Tracking Vehicular -type Transportation	■ Worker safety and goods retrieval efficiency in stock taking with forklift trucks [SI]; ■ Personalised shopping and store-level analytics through shopper or push cart tracking [169] [RE]; ■ User-friendliness with indoor car parking [170] [SA]; ■ Tracking large goods on a conveyor or assembly line for business intelligence [SI]; ■ the tracking of hospital medical (beds, carts) [HE]	■ Dependent on use case and/or moving velocity; ■ Medium to large coverage area; ■ Decimetre to metre-level accuracy in 2.5D; ■ Reasonable price; ■ Latency $(10 - 500$ ms) and update rate typically not that stringent	SPS	No dominant positioning technology; Compromising on prerequisite cost or accuracy; Cost associated with excess accuracy of LIDAR or ultrasound solutions hinders adoption; Low-cost (i.a. BLE) solutions too inaccurate; Even with stout UWB development, projection leaves room 'middle way 'next-gen technologies; IPS required for EMI-sensitive areas; UWB's adoption already being in progress is an abetment towards future market dominance	■ (u)VLP is well-grounded in vying for, depends on its cost-projection (i) to which extent are VLP-enabling measures, namely a LED driver retrofit and/or the addition of supplementary light sources, needed, and (ii) is VLP going to be endorsed and integrated as a service in illuminance networks; ■ Niche domains
Passive Tracking of Pedestrians or bi/quadripeds	■ Presence detection or occupancy sensing: –for moderating home and consumption [SA]; –for quality of service optimisation (i.a. handover) [6G]; –for location-dependent content access [6G]; –in geofencing (i.a. social distancing); -in perimeter sensing for the workers' well-being [SI,HE]; –for queueing density determination [HE]; –for sports performance analytics [SP]; –for emergency human detection [131]; –for personal or social shopping experiences [RE]; –for consumer or business intelligence [RE]; ■ (Health) monitoring: –in ambient assisted living [HE]; –in wireless body area networks [HE,SP]; –in livestock well-being [171] [AG]; ■ Location tracking: – for localising patients or on-call hospital staff [HE] –for staff (e.g., adept deployment) management [RE]	■ 3D submetre-order accuracy; ■ (Sub)second update rate; ■ Privacy guarantees	RPS; (Thermal) vision; WLAN; impulse radar UWB	Current IPS highly functional; UWB radar shows even more promise for passive tracking.	■ A more privacy-conserving method may entail sensing, and not imaging, changes in the visible channel, i.e. visible light sensing [12], [172], [173]; ■ By integrating a PD in each (unmodulated) light fixture, VLS can be effectuated with a small overhead. ■ sensing changes in the optical channel is a feasible and privacy-conserving method [172] ■ UWB radar shows even more promise; ■ Due its broadcast nature, VLP is only suited in downlink (from a gateway) sensor networks, but even there its energy-consumption renders it impractical.
Navigation or Active Tracking of Pedestrians or bi/quadripeds	■ Indoor (pedestrian) navigation [SA] –for patient's wayfinding, especially the visually-impaired [HE] –for shopping assistance (e.g., to desired products); ■ Location-dependent leisure i.a. augmented/ virtual reality [174] [SA] and tour-guidance [175] [SA]; ■ athlete tracking and body area networks [176]	■ Low to medium battery exertion; ■ (Sub)metre accuracy; ■ Robustness with receiver mobility in 6 DOF; ■ Profit from smartphone (integrated sensors) prevalence; ■ Significant coverage area; ■ Security service	WLAN for communication/ positioning, augmented with markers, BLE or UWB; Vision for precision tasks; UWB and BLE better fit active tracking	UWB's aptitude; The particularities of on/near body positioning, e.g., regarding the LOS link obstruction, deter directive IPS (such as IR); Smartphone's computational and sensory platform allow resorting to location-augmentation techniques, such as map matching for accuracy.	■ Delineated camera-based VLP's prospects; ■ tilt and LOS link obstruction associated with on/near body positioning; ■ not-quite-user-friendly nature of the compensation techniques hinder VLP's adoption; ■ Holding the smartphone-integrated receiver in the neighing-to-horizontal manner requested to either track the user's shadow or to capture a sufficient amount beacon signals feels unnatural, and unproductive. ■ When also considering the more extensive battery drainage, it is clear that VLP cannot be identified as the best solution from the mix.
Aerial Vehicle Navigation	■ Navigation of (miniature) UAVs [177]–[180]: –for efficacious automated inventory monitoring [SI]; –for packet handling; ■ indoor security or crop monitoring	■ Real-time ■ Centimetre-order ■ large coverage area ■ Highly robust ■ Lightweight ■ Energy efficient ■ reasonable price	Among which vision, optical flow; LIDAR are popular choice	Stringent QPoS only met with multiple IPS fusing; UWB regarded as, tried out as low-cost dominant IPS (to be fused with i.a. vision technology); Improvements needed	■ VLP could feature owing to its lightweight receiver attaining higher-than-UWB update rates. ■ remains to be evidenced to which extent the performance degrades as a consequence of drone flight's inherent significant receiver tilt and/or vibrations.

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TABLE IV

(Continued.) CLASSIFICATION OF LBS BASED ON THEIR ASSOCIATED QPOS AND IPS (SI: SMART INDUSTRY OR "INDUSTRY 4.0," SA: SMART ARCHITECTURES, RE: RETAIL 4.0, HE: HEALTH 4.0, AG: AGRICULTURE 4.0, 6G: SIXTH-GENERATION TECHNOLOGY, SP: SPORTS, RPS: REMOTE POSITIONING SYSTEM, WHERE INFRASTRUCTURE CALCULATES POSITION, SPS: SELF-POSITIONING SYSTEM, WHERE TAG CALCULATES POSITION

Type	Incentives (sector)	Important QPoS KPIs	Current IPS	Status and Developments	Perspectives for VLP
High-speed Vehicular Applications	■ Intelligent automotive transportation systems; ■ handover intermittently GPS-denied (tunnel) environments [181]; ■ vehicular-to-vehicular communications [182]–[184]; ■ vehicular ad hoc networks (VANETs); ■ autonomous car park	■ Capability for quick GPS-handover; ■ adequate update rate; ■ less imperative (sub)metre-order accuracy	Vision & LIDAR	IMU, Vision; With smartphone integration, GPS handover can also be envisioned for UWB performance depends on the tunnel environment, and the NLOS contributions it induces [185]	■ Opportunity in presence of visible lights, i.a. in the form of headlights/tail lights and illumination lamps in tunnels; receiver easily integrated next to the lights or behind the windscreen; ■ VL(C)P looks auspicious as an (addendum) IPS for vehicular ad hoc networks and intelligent transportation systems.
High Performance Applications	■ Robotic tasks during surgeries robotic surgery [HE]; ■ virtual reality entertainment	■ Real-time; ■ highly-available and robust; ■ cm (and even mm) order accuracy/ precision; ■ in 5/6 DOF (i.e. including pose estimation); ■ high availability [186]	(hybrid) infrared optical; (both inside-out/ outside-in tracking); ultrasound; mechanical [187]	With filtering and fusion, a subset of applications may not be too strenuous for UWB, when relaxed positioning requirement and a use for the communication aspect of UWB	■ May be too strenuous VLP, definitely for uVLP; ■ some entertainment-type applications suit VLP, with a relaxed positioning requirement not induce many LOS obstructions (otherwise more LEDs are needed) or a significant update rate and a milder accuracy relaxation.
Smart Ground-based (Autonomous) Vehicle Navigation	■ Automated guided vehicles (AGVs) – to form connected infrastructures [177], [188], [189] [14], [43] [SI], with automated goods monitoring and handling [RE]; – More humane picking robots or higher-yield precision planters in indoor agriculture [190] [AG]; ■ Guiding companion robots – for welcoming/assisting in child/elderly care [HE], – for shopping assistance [191] [RE] ■ Automatising cleaning robots [SI,RE]	■ Accuracy determines AGV safety margins, and allowable population density; ■ 10 cm upper bound aim in intelligent factories; ■ $20-30$ cm suffices when no exigency, in faithfully abiding by a predefined trajectory (e.g., a cleaning robot); ■ At least a location update every 100 ms	Via slotted wires; via (magnetic) guide tape or rails; via vision guidance; LIDAR, frequently in tandem with SLAM [192], [193]	■ Current approaches have restrictions: –No dynamic reconfigurability of AGV routes due to substantial relocation effort or frailty of guiding components; –Marker-based vision guidance is not low power; –LIDAR adds a substantial cost per vehicle and has its reflector infrastructure perceived as disturbing; ■ Need cheaper LIDAR alternative tolerating accuracy compromise especially with proprioceptive sensor fusion, e.g., using wheel encoders. ■ Applications trending towards employing UWB [188], [194], whose potential depends on application QPoS trade-off. ■ LIDAR-based when high accuracy is required; ■ its current QPoS metrics seemingly cause it to lose out to the ubiquitous LIDAR-based solutions, knowing that the latter is becoming more and more affordable. One reason lies in that certain multipath configurations induce significant p_{90}/p_{95} errors large AGV trajectory safety margin needs to be allocated, and a corresponding cost is incurred.	■ Supplementing the (sub)decimetre 2D positioning accuracy, destined to at least match UWB's, with a more profound cost differentiation, preordains minimal retrofitting VLP to give LIDAR a run for its money. When an adequate (illumination) coverage cannot be presupposed, VLP loses its appeal. Hence, VLP's potential largely revolves on the pre-existing roll-out and the latency, accuracy and cost trade-off of the specific application. With UWB/ VLP, fusing with among others vision data may frequently be required to reach the required accuracies. Even then, the accuracy curtailing prohibits dense robotic deployments.

Although less commonly used indicators, like robustness (when considered), are often treated in a qualitative way or are mentioned sideways, a quantitative formalization of each of these metrics would prove useful toward comparing different solutions. Then, a commonly used approach is the use of multiobjective functions that combine a (sub)set of quantitative QPoS indicators $\{\text{QPoS}_j\}$ into a fitness score, weighted based on their relevance to specific application needs, as follows [147], [148], [149]:

$$f_{\text{QPoS}} = \sum_{j=1}^{J} w_j \cdot \text{QPoS}_j. \tag{1}$$

Here, QPoS_j represents a quantified score with j referring to one specific indicator, such as accuracy, latency, and cost, and J the total amount of indicators considered in the optimization. The sign of w_j determines whether an increase in QPoS_j increases or decreases the fitness score, and the absolute value $|w_j|$ indicates the relative importance of the specific indicator QPoS_j to the total fitness score [149]. However, it is not always possible to rank two solutions, e.g., when they each perform better on different aspects (e.g., having a better accuracy but at a higher cost). Therefore, QPoS could also be formulated as a multiobjective Pareto optimization problem [150], [151], rather than through a fitness function [i.e., (1)]. While in the latter, all indicators are combined into a single scalar value f_{QPoS} that allows a total ordering of solutions [149], Pareto-based methods identify a multidimensional Pareto front, which represents a set of optimal solutions where it is not possible to make a single QPoS indicator better without making at least one other worse [150], [151].

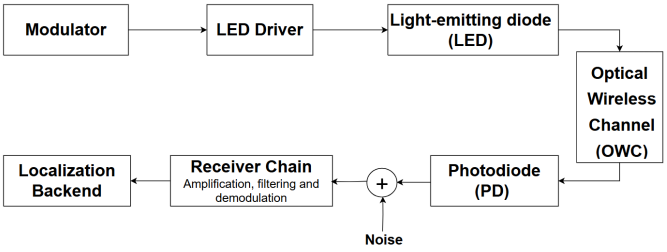


Fig. 5. Schematic representation of a typical VLP system.

In the VLP literature, system evaluation is often centered around accuracy, while other QPoS aspects listed in Table III are rarely addressed holistically [142]. However, a broader examination of QPoS indicators reveals that these metrics are influenced by design decisions throughout the entire VLP chain (see Fig. 5) [11]. For example, cost is affected by factors

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TABLE V
COMPARING THE DIFFERENT INDOOR LOCALIZATION TECHNOLOGIES, PART 1

Technology	Rationale	Drawbacks & Risk	Standards & Regulations	Technique	Scheme Operating Frequency
VLP	■ Concurrent illumination and Positioning, ■ Associated infrastructure reuse, ■ Not in RF spectrum, ■ Directed atto cell localisation	■ Illuminance compliance and health safeguarding e.g., glare, flicker, etc. ■ Expensive when no prior infrastructure or when luminous flux is lost. ■ Battery-based transmitter difficult ■ Ambient sunlight and LOS obstructions	IEEE 802.15.7/13, ITU-T G.vlc, IEEE 802.11.bb TG, etc.	RSS, AOA, & T(D)OA	Mostly $0.5-20$ kHz, though up to $5-20$ MHz or GHz if laser-based
uVLP	See VLP, and more: ■ Zero infrastructure cost, ■ Lights omnipresent in indoor environments	■ Illuminance Constraints ■ Time-variation CF and Limited SNR ■ Feasibility may be impacted with improved LED driver design	None	RSS	DC or $10-100$ kHz
(Acoustic) Sound (ACS)	■ (Phone) microphone reuse ■ Human speech frequently available ■ Remote positioning variant popular	■ Affected by (ambient) sound pollution, ■ Not low energy, ■ In audible range	No technology standard, Loudness well-regulated (Directive 2003/10/EC)	T(D)OA & AOA	$2-6$ kHz $15-20$ kHz (20 Hz - 20 kHz)
Ultrasound (ULS)	■ Potential for high accuracy ■ No electromagnetic interference (EMI)	■ At least specialized infrastructure ■ Limited positioning range and update rate ■ Not low energy ■ Accuracy-influence battery-level ■ Susceptible to RF crosstalk, near-far problem ■ Multipath reverberations/obstructions ■ (Self-)interference and sound range-leakage ■ Needs accounting for humidity & temperature	>20 kHz (safety) guidelines, no standards	T(D)OA -TWR AOA & PDOA	$20-60$ kHz (even $200-400$ kHz)
RFID	■ (Very) cheap and multi-purpose small form factor (passive) tags	■ Significant cost RFID reader ■ (Dense) deployment for adequate accuracy, but mutual electromagnetic coupling ■ (Tag) detuning, collisions, interference and data flooding. ■ Privacy and security concerns ■ Country and application-specific standards	ISO/IEC 18000/18092, ISO 11784/11785/14223 (animals), etc.	RSS, PDOA, (AOA & TDOA)	$125-134.2$ kHz $140-148.5$ kHz $13.553-13.567$ MHz $858-950$ MHz $2.4-2.483$ GHz, etc.
LPWAN	■ Long (battery) lifetime tag ■ Multi-building coverage	■ Inadequate performance indoors	Specifications and Regulations, (EU) 2019/1345 3GPP Release 13/14	RSS, T(D)OA, AOA	$433.05-434.79$ MHz $791-870/873$ MHz $880-960$ MHz $1710-1880$ MHz
WANET	■ Energy effective, ■ Compatible with multi-hop, wireless sensor networks	■ Inaccurate (interference) and not robust ■ Only meant for sensor networks	IEEE 802.15.4, IETF RFC 6282	**RSSi**, AOA, & TOA	2.4 GHz ISM band 868 MHz band
Bluetooth	■ Low energy, ubiquitous and cheap ■ IC integrated in phones	■ Prone to interference affecting scalability and update rate	Originally IEEE 802.15.1, Specification Bluetooth SIG Version 5.3	**RSSi**, A(D)OA, & DTDoA	2.4 GHz ISM band $(2.402-2.480$ GHz)
WLAN	■ Communication infrastructure reuse ■ Near-ubiquitous connectivity ■ Widespread ICs and phone integration ■ High throughput and NLOS feasibility	■ Too inaccurate/effort-heavy for certain applications, ■ Congestion and (self) interference, ■ Complex processing, ■ Dynamicity of the Radio map/environment	IEEE 802.11-2020 (ac/ax)	**CSI**, RSSi, AOA, & T(D)OA	$2.401-2.495$ GHz $5.15-5.895$ GHz $(5.925-7.125$ GHz)
UWB	■ Near-immunity, and limited contribution, to interference ■ Up to centimetre-level accuracy at limited cost, ■ Phone integration, ■ (Impulse radar's) high potential, ■ (CSI-based) multipath mitigation	■ Relatively expensive for eyed applications ■ Only recent market penetration ■ Update rate and scalability	IEEE 802.15.4 (a/z-2020)	**T(D)OA** -TWR, AOA	>500 MHz band in $3.1-10.6$ GHz
mmWave	■ Next-gen communication technology, ■ Small-size receiver, ■ Dynamic beamforming and Massive MIMO, ■ mmWave also for backhaul	■ Still in development phase, which means high cost, ■ Hardware complexity, ■ Energy consumption, range, coverage-aesthetics, considerations needed	3GPP Release 16 IEEE 802.11ad/ay	**T(D)OA** -TWR, AOA	$24-100/300$ GHz 3 GHz bands in $24.5-52.6$ GHz
LIDAR	■ High accuracy in practical environments, ■ Simultaneous localisation and mapping (SLAM), ■ Well-adopted for robotic navigation	■ High-operating cost, certainly for high-end LIDARs, ■ Limited update rate, ■ Computationally intensive, ■ LIDAR beacons perceived as ugly	No standard	TOA-SLAM	$\sim 193/(319)/331$ THz
IR	■ Variability in system configuration, ■ Potential for high-accuracy/precision, ■ Coherent and Directive non-visible LOS link, ■Availability remote-positioning variants	■ Ambient visible and infrared light interference ■ Difficult to ensure industrial-type coverage ■ Accurate IR bears a steep price tag ■ Regard human safety limits	InfraRed Data Association IEEE 802.11 (bb)	P(D)OA, AOA, TOA	$\sim 315-385$ THz ($\sim 100-215$ THz)
Vision	■ Rich location data (SLAM) ■ Markers low-cost and easily wielded ■ Widely-available camera (variants) ■ Phone-integration ■ Well-developed technology	■ Needs adequate light levels, ■ Too computationally intensive for constrained devices, ■ Standalone accuracy inadequate or too costly	IEEE 3333	AOA	$\sim 350/425-750$ THz

such as the modulation scheme (e.g., on-off keying (OOK) and code shift keying (CSK) [44]) by imposing requirements on LED bandwidth, and receiver complexity for demodulation [153], [154], [155], by PD sensitivity, LED power and density [156], and the extent of prior system calibration required [157]. Computational complexity can be heavily influenced by the employed channel modeling, where phenomena such as reflection can be considered for improved positioning [158]. Latency and computational load are further shaped by backend signal processing algorithms used for demodulation (e.g., fast Fourier transform (FFT) [159] and multiple signal classification (MUSIC) [70]) and localization, which can either streamline or burden real-time positioning performance [156]. Robustness is influenced by the LED density, LED stability, external lighting conditions, and the deployed algorithm [142], [156]. Despite this, the impact of transmitter, channel, and receiver components on QPoS is often underexplored in the VLP literature, with most research focusing on accuracy at the

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localization backend. Typical assumptions include common transmitter configurations [e.g., four Lambertian emitters in a square arrangement using frequency division multiple access (FDMA)], an LoS channel, and a single PD receiver. It is clear that the set of QPoS indicators forms a complex interplay, which becomes particularly important when transitioning from academic evaluations to commercial deployments.

IV. VLP in the IPS Landscape

This section will now take a broader view on things. First, it will consider different positioning application types and discuss VLP's perspectives toward adoption. Then, VLP will be better situated among its indoor self-positioning technology competitors, by benchmarking VLP and other IPS technologies, considering different relevant QPoS aspects. Although obviously not straightforward, this work attempts to provide a comprehensive overview.

A. (Future) Indoor Applications

Fig. 4's rendition of the current IPS landscape accentuates that (u)VLP may cover a part of the spectrum that preexisting technologies are not able to. There, (u)VLP is able to supply more accurate or more economical positioning than the current dominant competitor(s). Moreover, it is projected to shift further toward the origin of Fig. 4.

To more easily overview which types of indoor LBS for which VLP could be (or already is) suited, the LBS are assorted into groups of similar application types, each with a matching QPoS requirement [119], in a manner that is irrespective of their overarching sector. Table IV shows this application-based categorisation. Per the applications class, the dominant primary localization system(s) are given, in conjunction with an assessment on whether better performing IPS are a future requisite and whether VLP poses a solution.

B. Benchmarking (Indirect) Self-Positioning Technologies

Unfortunately, diligently benchmarking IPS technologies solely based on literature is onerous [193], [194], [195]. The reasons being the following.

1) An IPS technology features in many system configurations, e.g., with different transceiver types and calibration requirements.

2) The reported localization results inherently have various dependencies. The irreproducible roll-out, the localization parameters, and the employed result metric [118] all play a role.

3) These positioning data generally have not been collected in an application-representing scenario. A sparse data collection in an idealized small lab setup is common [196]. Indoor positioning competitions definitely aid in appraising IPS [197], [198]. Unfortunately, these have yet to incorporate newer technologies, such as VLP.

4) Vital (calibration) parameters and QPoS KPIs (e.g., energy consumption) tend to not be disclosed.

5) Finally, as sensor fusion is frequently used to ameliorate the positioning, the error contributions of a single IPS are not easily sifted out.

Nevertheless, in a survey of the QPoS indicators, anno 2024 is instrumental in highlighting the disparity between IPS technologies. Hence, at the risk of quickly becoming antiquated, Tables V–VI present a subjective author-valuation based on an extrapolation of results from an extensive literature search [185], [199], [200], [201], [202], of: audible sound (ACS) and ultrasound (ULS) [203], [204], [205], [206], [207], [208], [209], [210], [211], [212], [213], [214], [215], [216], [217], [218], [219], [220], [221], [222], [223], [224], [225], [226], [227], [228], [229], [230], [231], [232], [233], [234], [235], [236], [237], [238], [239], [240], [241], [242], [243], [244], [245], [246], [247], [248], [249], [250], RFID [251], [252], [253], [254], [255], [256], [257], [258], [259], [260], [261], [262], [263], [264], [265], [266], [267], [268], [269], [270], [271], [272], [273], [274], [275], [276], [277], [278], [279], [280], [281], [282], LPWAN [283], [284], [285], [286], [287], [288], [289], [290], [291], [292], [293], [294], [295], wireless ad hoc network (WANET) [296], [297], [298], [299], [300], [301], [302], [303], [304], [305], [306], [307], [308], [309], [310], [311], [312], [313], [314], [315], [316], [317], [318], Bluetooth [319], [320], [321], [202], [322], [323], [324], [325], [326], [327], [328], [329], [330], [331], [332], [333], [334], [335], [336], [337], [338], [339], [340], WLAN [341], [342], [343], [344], [345], [346], [347], [348], [349], [350], [351], [352], [353], [354], [355], [356], [357], [358], [359], [360], [361], [362], [363], [364], [365], [366], [367], [368], [369], [370], [371], [372], [373], [374], [375], [376], [377], [378], [379], [380], [381], [382], [383], [384], [385], [386], UWB [121], [122], [123], [126], [160], [161], [162], [163], [183], [192], [387], [388], [389], [390], [391], [392], [393], [394], [395], [396], [397], [398], [399], [400], [401], [402], [403], [404], [405], [406], [407], [408], [409], [410], [411], [412], [413], [414], [415], [416], [417], [418], [419], [420], [421], [422], [423], [424], [425], [426], [427], [428], [429], [430], [431], [432], [433], mmWave [434], [435], [436], [437], [438], [439], [440], [441], [442], [443], [444], [445], [446], [447], [448], [449], [450], [451], [452], [453], [454], [455], [456], [457], [458], [459], LIDAR [460], [461], [462], [463], [464], [465], [466], [467], [468], [469], [470], [471], [472], [473], [474], [475], [476], [477], IR [478], [479], [480], [481], [482], [483], [484], [485], [486], [487], [488], [489], [490], [491], [492], [493], [494], and vision [495], [496], [497], [498], [499], [500], [501], [502], [503]. VLP's table entries are completed on the basis of QPoS.

In view of the QPoS discrepancy found within the same IPS technology, Tables V–VI appraise the typically deployed and generally most economically scalable variant(s). For simplicity, their economic aspects only include the hardware costs. Only indications are given for the other cost structures, regarding, e.g., installation, maintenance, or operations. Moreover, in compiling the VLP-competitors, only systems were considered that are able to provide stable and absolute one-shot indoor localization, i.e., with a limited drift over time. Technologies effectively omitted are, i.e., dead reckoning (e.g., based on IMU data), magnetic positioning, image-based technologies operating on visual features (e.g., optical flow), and GNSS. Finally, it needs to be remarked that most contemporary IPS

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TABLE VI
COMPARING THE DIFFERENT INDOOR LOCALIZATION TECHNOLOGIES, PART 2

Technology	(Projected) Performance				Scalability & Robustness	Energy efficiency of receiver
	Typical p_{75}	Harsher p_{75}	(Practical) Reception Range	Refresh rate & Latency		
VLP	$3-15$ cm	$10-40$ cm	$6-10$ m (15 m)	>100 Hz 50 ms	■ Broadcasting configuration inherently very receiver-side scalable; ■ Issue of LOS obstructions and receiver tilt	Unoptimised: 100 mW (max. 5 W)
uVLP	$10-50$ cm ($5-20$ cm at low rate)	>100 cm	$3-5$ m (6 m)	$1-10$ Hz ($100-200$ ms)	■ Broadcasting configuration very scalable; ■ Issue of LOS obstructions and receiver tilt; ■ Robustness to time-variation needs working on	Unoptimised: 100 mW (max. 5 W)
Acoustic Sound (ACS)	$\sim$0.5 m	$1.5-2$ m	$7.5-12.5$ m (50 m)	$\sim$1 Hz ($50-100$ ms)	■ Limited due to crosstalk or ambient sound	$0.1-1$ W
Ultrasound ULS	$0.3-5$ cm	$10-30$ cm	$5-10$ m (25 m)	$\sim$1 Hz (max.16 Hz) ($5.9-250$ ms)	■ Limited scalability and robustness due to (self) interference, low update rate and degrading impact receiver/ transmitter tilt	$0.1-1$ W
RFID	$1-5$ m (active), 2D: $\sim$30 cm, when tags very dense	$1-10$ m	~ 10 cm $2-8$ m $30-100$ m active	$\sim$1 Hz ($<<100$ Hz) Up to 7.5 s	■ (Tag) collisions, interference and data flooding	<50 mW (tag) $\sim 0.3-15$ W (reader)
LPWAN	$\sim 2-4$ m	$\sim 3-20$ m	$\sim 5-15$ km (>700 km)	$<<1$ Hz	■ Largely-scalable, not robust	$\sim$225 mW (Rx) $\sim$775 mW (Tx)
WANET	$\sim 1.5-4$ m	$5-10$ m	$5-10$ m (50 m)	<1 Hz	■ Scalable, not robust	$10-100$ mW
BLE	50 cm-$1-3$ m	$0.5-2/5$ m	$10-15$ m (50 m)	$1-3$ Hz ($0.2-30$ Hz) $50-300$ ms	■ Supports approx. 7 tags per anchor; ■ Interference increases with device count	$10-100$ mW (400 mW)
WLAN	$1.5-4$ m ~ 50 cm, special deployment	$3-10$ m	$25-50$ m (100 m)	$0.1-10$ Hz $0.5-3$ s	■ Scalability limited due to interference ■ 30-some tags supported per cell ■ Wide coverage mitigates failures ■ Performance impact by channel time-variability	~ 1 W >0.5 W (Tx) >0.2 W (Rx)
UWB	$3-15$ cm	$10-40$ cm	50 m (>100 m)	$10-100$ Hz $50-100$ ms	■ 10-30+ tags supported per cell, depending on positioning scheme and update requirement; ■ Resolving multipath components adds robustness	<10 W (Tx) possible <20 mW (Rx) possible max. 0.7 W
mmWave	$0.15-2$ m	$1-4$ m	$0.5-2$ m ($5-10$ m)	Not reported $40-60$ ms	■ Scalability limited due to interference ■ Massive MIMO/beamforming serves many concurrent users	$\sim$5 W
LIDAR	$0.5-2.5$ cm	$2.5-5$ cm	75 m ($100/250$ m)	$0.3-3$ Hz (max. 20 Hz) $0.05-1$ s	■ Disregarding cost, large scalability potential	$\sim$10 W
IR	$0.1-5$ cm $15-35$ cm (depending on type)	5 cm $0.5-2$ m	8 m (15 m)	$0.2-30$ Hz $5-50/500$ ms	■ Scalability depends on system-type from 5-10 (camera) to 100+ (PD); ■ Mainly tilt and LOS obstruction hinder performance	>2 W (infrared light) 0 W (marker) $\sim$50 mW (PD) $\sim$10 W (camera)
Vision	$1-5$ m ~ 50 cm, when dense	$5-10$ m	2 m (10 m)	$30-120/240$ Hz $10-455$ ms detect high performance event camera	■ Large-scale deployments possible ■ Imaging artefacts prohibit robustness	>10 W (performance) (As low as $\sim$0.5 W)

still have accuracy boosting, cost-reducing, and robustness increasing measures still on the horizon.

Tables V–VI provide the rationale for both the historically grown IPS fragmentation and the continued demand for IPS development to (better) cater to existing and future LBS. After all, each IPS has an inherent, proprietary QPoS tradeoff. As such, neither IPS can be identified as "the most optimal." There is, to date, no single IPS that is capable of seamlessly providing a high level of accuracy at a very low cost and a low energy consumption, whilst preserving privacy and security [506]. Hence, an IPS's aptitude will need to be considered on an application basis. As an IPS' QPoS, and especially its cost and its accuracy/precision, is highly dependent on its roll-out, the latter is specifically tailored to the application at hand.

V. FUTURE PERSPECTIVES

This penultimate section briefly encompasses our views to venture beyond both the IPS field's and VLP's status quo. It collects VLP's open problems, and proposes future research directions, which all have the potential to improve VLP's QPoS aspects. As a general overview, Table VII summarizes possible improvements to the QPoS KPIs that were mentioned before, touching upon some research opportunities for VLP.

A. Current Research Trends in VLP

1) Machine Learning: MBF-type of algorithms can be (too) cumbersome, in terms of complexity, execution time, and storage, for large-scale environments, even when using sparse models and fast location matching methods. With the truthful propagation models employed for VLP, ML, or specifically, deep learning is turned to. Data augmentation methods, with e.g., a specified noise distribution, installation errors, or radiation pattern divergences, allow training a more robust localization model. Tilt-agnostic VLP may have to heavily rely on neural networks as well, either for ranging correction or direct localization. Another possibility for further investigation is focusing on the use of ML to overcome the challenges given by LED blockages [507], [508], which has shown great

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TABLE VII
COMPARING THE DIFFERENT INDOOR LOCALIZATION TECHNOLOGIES, PART 3

Technology	Commercial		Privacy, Safety & Security	Deployments, Companies & Products
	(Projected) Cost per tracker	(Projected) Cost Infrastructure (per 25 m^2 of large-scale area, not including manual labour)		
VLP	€75 (reduces with scaling)	€0/50/400 depending on prior availability	■ In its simplest form no encryption ■ Typical signal confinedness and link distance require eavesdroppers to be in view ■ Resistance against DDOS	■ Targeting VLC: Signify, pureLiFi Ltd., Oledcomm, General Electric, Nakagawa Laboratories Inc., VLNComm Inc., LVX System, Outstanding Technology, Velmenni Technology Co. Ltd.
uVLP	€75 (reduces with scaling)	€0	■ No encryption ■ Inherent 'security' with light propagation	■ None on record
Acoustic Sound (ACS)	€5-30	€80 €1500	■ Encryption complexity is constrained ■ Pick up of non-signal auditory information enables deanonymization attacks ■ See under 'Ultrasound'	■ Siemens ■ Beep [243]
Ultrasound ULS	€15-100/400	€800 €2000+	■ Room-confinedness entails security ■ Vulnerable to denial-of-service (DOS), jamming and man-in-the-middle attacks (impersonation) ■ See under audible 'sound'	■ Marvelmind robotics, Sonitor technologies AS, Sonarax Technologies, Televic Group ■ (SNoW) BAT, Cricket, TELIAMADE, Silent Whistle, LOSNUS, Lok8, etc.
RFID	€2-10 €500 + (up to €20000)	€50 €2000+	■ Only seldom encrypted ■ Security threats in discoverability, replay, snooping, eavesdropping, skimming, tampering and cloning	■ Keonn Technologies, MetraLabs GmbH, Aucxis, Fetch Robotics Inc., Zebra Technologies, LANDMARC - RFID
LPWAN	€8-40	∼€0.1 (but large upfront investment)	■ Authentication and encryption supported ■ Replay attack may be unnoticed ■ Easily jammed	■ Semtech Corporation, Loriot, u-blox, Aloxy, Orange, Vodafone, infsoft, LoRa Alliance®
WANET	€5-30	€150	■ Information leakage ■ Vulnerable to DOS, jamming	■ Aloxy, Texas Instruments, ZigBee Alliance, etc.
BLE	€5-30 €0 (phone reuse)	€120 beacon	■ Recent security measures ■ Discoverability and impersonation potential ■ Bluesnarfing, (misplaced) trust in paired device	■ Quuppa Oy, BlooLoc N.V., BlueIOT, Locatify, u-blox, Esri (Indoo.rs), Link Labs (AirFinder), Samsung Electronics
WLAN	€4-50+ €0 (phone reuse)	€100-1000, or infrastructure-reuse	■ Encryption possible ■ Threatened by DOS attack, jamming rogue access points, etc. ■ Not for EMI-sensitive environments	■ Navizon Inc., infsoft, Cisco, Inpixon, Liberty Broadband - Skyhook, Ekahau, Horus, SpotFi, Phaser
UWB	€6+ (IC) €35 (Finding) €100-400+ (Positioning)	€100/150	■ Resilience against jamming ■ Low probability of detection and interception ■ Safeguards needed against man-in-the-middle attack ■ Privacy measures needed for (impulse radar) sensing, w.r.t. heart/respiration monitoring and gait recognition [131], [507], [508]	■ Qorvo (Decawave), Pozyx, NXP Semiconductors, Ubisense, Lopos, Time Domain Corporation®, STMicroelectronics N.V. (Bespoon), Apple Inc.
mmWave	€10 (€4) for ICs, e.g., Qualcomm QCA64x8	€360-2400 (e.g., 2x Nighthawk XR700) €3000+	■ Vulnerable to jamming and nomadic beam-stealing	■ No 'solutions' offered ■ Transceiver equipment: e.g., Rohde & Schwarz, Texas Instruments
LIDAR	∼€300-400 (low-end) ∼€2000-15000 (high-end)	€30 (beacons cheap reflectors) €0 (when infrastructureless)	■ DOS with IR burst ■ Spoofable by rogue reflector beacons ■ Limited through-obstacle propagation entails privacy	■ Ningbo Osight Photoelectric Technology Co. Ltd., YellowScan, SICK Sensor Intelligence, Velodyne Lidar, Robert Bosch GmbH, Luminar Technologies Inc., Garmin - PulsedLight, Intel (RealSense)
IR	€3-30+ ∼ €0 €600+	∼€15-200+ €0 €1200+ (even €10000+)	■ Generally safe, still regard human safety ■ DOS via link obstruction ■ Limited through-obstacle propagation entails privacy	■ Ps-Tech, Vicon Motion Systems Ltd. ■ Active Badge, Firefly, OPTOTRAK, IRIS LPS
Vision	€150-1000	€0-5 for markers, not including placement effort	■ Encryption possible ■ Potentially infringing others' privacy ■ Easily spoofed and even DOS attacked (added tags) ■ All-environment safe	■ Nanchang IKV Robot Co. Ltd., Cognex Corporation, etc.

potential when used together with additional signals such as magnetic fields [509] or IMU sensors [510], [511], [512].

2) Solar Panel: The simultaneous powering and localization of a mini-drone with a (self-reverse biased) solar panel presents an exhilarating application for (u)VLP, where the solar panel serves at the same time as photosensitive receiver and as energy source for the VLP receiver module. Their bandwidth supports demultiplexing of LED signals at different frequencies, and under normal lighting, the SNR is sufficient for reasonable accuracy [513], [514]. Further research into large-scale deployment, solar cell types, multireceiver setups, and energy budgets could advance battery-less VLP, ideal for low illumination environments [515].

3) Hybrid IPSs: A hybrid IPS, combining VLP with UWB, event-based vision, IMU-based pedestrian dead reckoning, is expected to deliver an improved QPoS, but should at the same time allow for a seamless handover between technologies. With differing typical layouts and error profiles, positioning benefits seem plausible, with fusion with IMU-derived features currently the most researched avenue, increasing VLP robustness at a relatively low additional cost. It is an important study area, in which the set of sensors is typically tailored to

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TABLE VIII
IMPROVEMENTS FOR THE QPOS KPIS OF THIS ARTICLE'S VLP SYSTEMS

QPoS Factor	Methods for improving
Accuracy and Robustness	■ Offer reliability indicator based on SNR estimate, photocurrent, and position estimate. ■ Integrate mitigation algorithms in one, encompassing the tracking algorithm.
Compatibility	■ Dimming: introduce guard 'time frames' in which no modulation is used. ■ Illumination: employ practical application-ready LED lamps, with driving current that maximises lifetime and typical methods to avoid glare and dissipate heat.
Computational Complexity	■ Hybrid localisation algorithms, or embedded ML instead of unoptimised MBF. ■ Exploit the 'cellular' nature of VLP roll-outs. ■ Optimise map information and location searches, incorporating tracking information.
Cost	■ Use a more cost-effective receiver, e.g., a cheaper PD. ■ LED strip-like topology to reduce the number of modulators.
Delay	■ Limit computational complexity for fast search. ■ Code optimisation. ■ Local (on-board) processing of photocurrent.
Power consumption	■ Decrease modulation depth via transmit power scaling (supporting 'Darklight' operation [126]). ■ Introduce receiver idleness periods during guard 'time frames' with soft-synchronisation (IEEE 802.15.7 PHY [521]). ■ Introduce solar panel for constrained devices.
Privacy and Security	■ Restrict FOV of LED and PD with an optical concentrator. ■ Introduce codes in waveform (CDMA) or frequency hopping using near-synchronisation between LEDs.
Required Calibration	■ Autocalibration via floor reflection. ■ Switch to RRSS (or AOA) to reduce calibration effort. ■ Geometrical localisation.
Transmitter Scalability	■ Identify correct positioning cell among frequency reuse. ■ Optimise modulation frequency allocation. ■ Provide OFDM for dense deployments.
Update rate	■ Utilise larger modulation frequencies via improved COB LED driver or individually addressing sub-LEDs.

the (engineering) budget of the targeted application. A niche study may entail i) combined (u)VLP and UWB localization and ii) fusing various IPS with the cost-effective (L)SOOP approach of uVLP. From the VLP standpoint in VLP-UWB fusion, UWB could function as a robustness fallback in case of significant receiver tilt or LOS blockage, as a communication means, for LED location and $P_{t,i}$ calibration when integrated in a lamp, as to aid with indoor/outdoor transitions. From the UWB standpoint, i.e., NLoS classification and mitigation could benefit from (u)VLP, providing scalability and at least zone-level accuracy. Moreover, combining (u)VLP and UWB would allow optimizing the anchor density and placement, depending on what is available. Nonetheless, the question remains in which cases the hybrid IPS makes economical sense.

4) Visible Light Sensing (VLS): Due to a huge number of applications, we are already noticing extreme pressure on the 5G networks' available spectrum, which is encouraging the development of the 6G network. Joint indoor communication and sensing are essential in this new network. The flexibility to adapt VLP systems for joint communication and VLS allows it to satisfy this demand [516] and renders VLS research more alluring [171]. According to [12], while gesture recognition, occupancy detection, and passive target localization (and tracking) have received considerable attention in VLS, many exciting possibilities remain largely unexplored. As a passive and energy-efficient sensing method, VLS is uniquely positioned to meet 6G's needs for low cost, low complexity, lower battery consumption, device-free operation, and ultralow latency [12], [170], [171].

5) Location Tracking for VLP: One-shot VLP is accuracy-deficient and often not robust in the industrial environment. In the industrial environment, location tracking using (stateful) filtering was shown to be beneficial, often in combination with (IMU) sensor data. The degree of robustness that tracking adds against shadowing or multipath-induced outliers, small receiver tilts, LED installation errors, etc., is insufficiently characterized as of now. A guide is needed on which tracking algorithm is best used, which gain can be effectuated for the typically present noise profiles and the required update/mobility rates. Conversely, the LED sparsity level may also be enlarged when using location tracking algorithms.

6) Receiver and Transmitter Tilt: For receiver tilt compensation, embedded ML (actually deep learning) techniques can be used instead of relying on an IMU [58]. Utilizing a VLP receiver that consists of a single PD with a nonzero angle of inclination or that is based on multiple PDs [86] will allow unambiguously determining a receiver's (e.g., a drone's) yaw (with respect to the landing platform). Transmitter tilt estimation can also rely on ML-based approaches or compensate via a propagation model recomputation based on sparse measurements.

B. Recommendations and Opportunities for Research and Development on VLP

Apparent from QPoS is that literature's VLP systems are neither product-ready and nor able to meet the predefined QPoS targets of Section IV-A. Also, the larger scale evaluations described in Section II-D have shown that there are several challenges that the nascent VLP needs to address. In response, the following research directions and/or recommendations are formulated as follows.

1) General:

1) **More Extensive 3-D Accuracy Evaluations and Algorithms:** The industrial VLP experiments should be extended, first to 2.5-D and then 3-D localization. After all, the industrial environment will feature more drones, whether deployed autonomously or in tandem with a carrier AGV. In 2.5-D, a helper height sensor or prior location information is used.
2) **Standardization in VLP:** As the illumination market undergoes rapid transformation, it is crucial for VLP to pursue a QPoS that is ready for adoption. A key part of this process involves identifying a set of standardized VLP transceiver modules and system principles [26], [44], [517], e.g., a widespread, configurable VLP receiver that is able to account for the large swing in SNR, e.g., found with drones. In the short term, VLP's economic success depends significantly on its integration into lamps. With VLP still actively developing, it should aim to play a larger role in existing but underutilized standards like IEEE 802.15.7 [518] and ITU-T G.vlc [519]. The integration of Li-Fi by the IEEE 802.11.bb Task Group within the Wi-Fi ecosystem also presents an opportunity to build upon [130], [520].
3) **Applicability of Unmodulated VLP:** Depending on the required application and its QPoS, the illumination infrastructure can then be dynamically reconfigured. When no visual task needs to be performed, the

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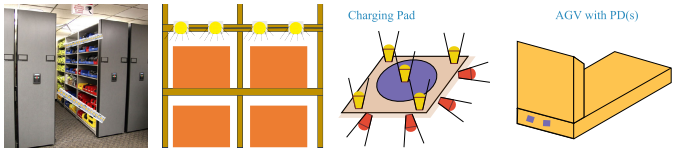
"darklight" principle can be applied and energy can be preserved. Otherwise, depending on the visual task, a switch can be made between VLP (e.g., for robots) and uVLP (e.g., for humans) [10].

4) **Open-Source VLP Simulator:** For system design, testing, and benchmarking, comprising the entire VLP chain (transmitter types, propagation models, receiver types and configurations, and positioning algorithms) in a modular way. Building on simulator tools initiatives, the simulator would ideally also permit simulating dynamic deployments with, i.e., tilting and/or obstructing objects.
5) **Li-Fi:** The ability of switching between intensity levels at a very fast rate, which is imperceptible to humans, is a key-enabler of VL(C)P. Importantly, it enables data transmission to be used for communication/positioning, whilst concurrently maintaining the lamp's original illumination purpose. Moreover, Li-Fi can be integrated to a backbone network via, i.e., power over Ethernet (POE) or powerline communication (PLC). It can obtain and benefit from additional UE position and orientation information.
6) **Sensor Fusion:** In 2-D/3-D VLP systems, relying solely on light signals can be inadequate, particularly when the signal is obstructed, or the receiving device is held in different orientations [512], [521]. For instance, slight tilts of a smartphone or PD during movement can introduce positioning errors that are difficult to mitigate using VLP alone [512]. To address these challenges and enhance positioning accuracy, sensor fusion is often employed by integrating additional components such as IMUs [510], [511], [512], lasers (e.g., ray-surface positioning) [521], [522], [523], RF modules [524], and extra PDs [21], [86]. These peripherals provide complementary information, such as receiver orientation, movements, and link distance, thereby enhancing system robustness [521]. However, the added cost and complexity of these components must be justified by the performance gains they deliver [521].

2) Transmitter:

1) **VLP-Compatible Transmitter:** For standalone VLP, perhaps the most pressing need is a transmitter that is illumination-compatible and has an appropriate LED driver. It should ensure a coexistence with its primary illumination function, specifically, where (PWM) dimming was not provisioned by decreasing the overall modulation loss. Utilizing restricted illuminance loss modulation, it may operate efficiently at both low (e.g., darklight principle) and minimal supported modulation or maximal illuminance (uVLP) levels [10]. When VLP is readily supported, a transmitter network combining all three functionalities: "regular VLP," "uVLP," and "darklight" operation [124] has merit.
2) **Driver and/or Modulator Count Reduction Mechanisms:** These have the potential to reduce VLP's price tag. The example solutions of [67] and [195] for square waves-based SC FDMA, employ a frequency halver circuit to sequentially derive the modulation frequencies of the individual LEDs in a strip, or of the individual LED strips in a lamp, from a common frequency synthesizer module.
3) **Study Into Transmitter Infrastructure Optimality:** This is required to determine what anchor density of what transmitter type, i.e., in physical dimensions and radiation pattern terms, best suits a roll-out, by trading-off QPoS and illumination metrics. Definitive answers are required regarding "Is it more interesting to add a VLP-dedicated point source to, or rather to (partly) retrofit, the nonpoint source lamp?," "What percentage of lamps needs to be VLP-enabled?," and "What is the optimal driving current scheme?" Swapping the visible LEDs for their usually more directive IR counterparts, or for laser diodes, may benefit the range and RSS gradient of the system, while the LED strip-based separation into the individual transmitter contributions presents the opportunity of a more fine-grained positioning algorithm modification. Ideally, this would lead to an industry-ready VLP network planner for predicting the illuminance and QPoS, and performing automatic design of a prospective VLP roll-out [76], [525].
4) **Protocols for Convenient Calibration and Deployment:** For a wide adoption of VLP, facilitating installation, minimizing on-site calibration, e.g., through auto calibration, and dealing well with, i.e., opaque and mirroring surfaces, and with lamp degradation (e.g., aging) is important. Standardized transmitters and/or receivers are desirable in this respect.
5) **Intricate Transmitter Deployments:** Dealing with, e.g., a single row of non-Lambertian and/or nonpoint source-like lamps found in hallways or in view inside warehouse storage racks passages, enlarges VLP's application potential. Particularly, non-AoA systems will require specialized lamps, lamp modifications, or other solutions to localize in the transversal dimension, such as asymmetric radiation patterns (e.g., with lens sheets), adding symmetry-breaking transmitters, decomposing composition lights into multiple transmitters, and sensor fusion and/or location tracking. color-division through exploiting optical birefringence with scotch tape [526], [527], or the modulation of the separate inner LED strips [195] can be identified as lamp decomposition solutions. In [195], square waves-based SC FDMA is effectuated by sequentially modulating the left and right half of a commercial-available linear LED lamp that consists of multiple LED strips.
6) **Device-to-Device and Device-to-Datasheet Sensitivity:** It is insufficiently reported how transmitter characteristics vary with respect to the datasheet or between transmitters. It is also advisable that device manufacturers report more extensive device characterization data, e.g., detailed radiation patterns. If necessary, what QPoS degradation-coping methods can be devised? Although mainly applicable to VLP transmitters, the same goes for VLP receiver devices.
7) **Light Exposure:** A less investigated research topic is the effect of LED lighting and modulation on living organisms. In humans, flickering was shown to have biological

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(a) (b) (c)

Fig. 6. Wielding VLP for (a) safe autonomic drone navigation through tightly spaced storage racks, (b) both accurate product localization and stable drone hovering, and (c) last meter AGV navigation as part of a future mobile charging solution. (c) Moveable wireless charging pad equipped with two LED strip subsystems to guide an AGV with multiple PD-based receiver(s) to its center (purple). Incorporated in the pad's side, the first strip (red) serves to navigate the AGV onto the pad. Once there, fine-grained alignment is ensured with a second LED strip (yellow), whose LEDs roughly pointing upward help a PD-based receiver at the AGV's bottom self-localize.

effects such as headaches and epilepsy, leading to the assumption of a minimal frequency of 200 Hz as being a safe low bound for modulation. While humans generally do not visually perceive flicker above 100 Hz, other living species, such as bees, have a so-called critical fusion frequency that is well higher, up to as high as 400 Hz [528]. Flickering light was also shown to reduce the number of attracted nocturnal insects [529]. More research into actual biological effects is warranted.

8) **Photonics:** The symbiosis with the field of photonics will support continued innovation for VL(C)P. The adoption of quantum dots, the exploitation of all signage, lighting, and display equipment enabling a ubiquitous positioning service, and the fiber-to-the-lights paradigm, in which a high-speed light module directly transmits the optical fiber's signals, are all thrilling examples.

3) Receiver:

1) **Novel Receiver Devices:** Manufacturing processes for 2.5-D and 3-D multi-PD receiver structures and their experimental evaluation (amount, position, and orientation of PDs) may be considered as an open challenge direction with VLP. Most available research is restricted to simulations.
2) **Luminaire-Based AoA and NLoS:** AoA VLP is typically considered the most promising PD-based VLP method. However, its positioning also degrades in the neighborhood of obstacles and nonpoint-source LEDs, for which NLoS compensation measures can be designed. Rather than to tackle the latter on the level of the channel model, aperture-based receiver structures can be employed [93]. Whether such an extended receiver is worth the extra complexity still needs to be determined, in conjunction with arbitrary radiation patterns and with uVLP operation and its lower SNR [10].

4) Applications:

1) **Precision Drone Landing or AGV Arrival:** Its principle relies on the integration of a cost-effective string of LEDs into the drone landing zone. As these LEDs can sequentially be strung together to form an LED strip, the resulting VLP system can limit its price tag. A similar system would enable precision drone flight [see Fig. 6(a) and (b)] [195]. Another "small-scale VLP" application comprises the last-meter navigation of AGVs to their ground-based wireless charging pads, similar to the drone landing platform, as shown in Fig. 6(c). With these systems featuring in the concept and early prototype stage, aptitude assessments are still awaited.
2) **VLP in Automotive Tunnel and Underground Parking:** Armature-based VLP is ready for implementation in a tunnel environment, where conditional on the receiver velocity either proximity or full-fledged localization can be turned to. Note that for higher velocities, another aspect worthwhile investigating is the tradeoff between positioning latency and accuracy, whereby the latter is improved by averaging out consecutive measurements for a higher SNR. The same principle could be used for navigation in underground parking, while the VLS counterpart could be used as an approach for monitoring traffic density or detecting free parking spots.
3) **1-D VLP:** Mostly overshadowed by 2-D and 3-D applications, 1-D VLP has application potential as well. For example, determining the longitudinal positioning of a robot along a guide (rail) materializes in the form of a picking robot in indoor agriculture. Based on the RSS gradient and with (structural) differential RSS (DRSS) or ratio RSS (RRSS)-based localization, there, a fine positioning granularity can be obtained. In a warehouse setting with high storage racks, the hallway is typically only covered with a single row of LED fixtures. With regular 1-D VLP, at least the longitudinal dimension is served with adequate positioning. localizing in the transversal dimension then requires a helper sensor and/or other lamp modifications (see earlier "intricate transmitter deployments").

C. Guidelines for VLP Evaluation and Reporting

To improve reproducibility and enable fair benchmarking across VLP systems, we propose a concrete evaluation protocol and reporting guidelines for authors to follow. Studies should report detailed emitter specifications, including LED type, radiation pattern, emitted optical power, modulation scheme, installation locations and orientation, as well as receiver specifications such as PD type, active area, installation height and orientation, responsivity, field of view, and noise figure. The experimental environment must be thoroughly described, including room dimensions, surface materials and reflectivities, and ambient illumination conditions. Authors should document calibration procedures for LED radiation patterns and PD acceptance models. Measurement locations or trajectories should be spatially referenced with respect to the transmitter configuration. Sampling rate and preprocessing steps, such as averaging, should also be specified. Ground-truth acquisition methods must detail the system used, its accuracy and resolution, and how temporal and spatial synchronization with VLP data was ensured. Evaluation results should report localization errors (e.g., 75th and 95th percentiles), availability, and—where relevant—end-to-end latency. Costs should be summarized using bill-of-materials (BOM) estimates, with dominant components identified.

For data-driven methods (i.e., ML-based approaches), reproducibility requires releasing raw measurements, processing and positioning code, and explicit train/validation/test splits.

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Existing open datasets, such as the Autonomous Fingerprinting dataset [530], E-VLC [531], OWP-IMU [532], and QUAPOS [533], along with reporting templates like ORDIP [534] developed for indoor positioning, provide practical references and illustrate the benefits of standardized, high-resolution VLP data. By following these guidelines, future studies can produce comparable, reproducible, and actionable results that facilitate rapid progress in VLP research.

VI. Conclusion

An ever-growing demand for location-based services (LBS) drives the continued development of IPSs to meet the increasingly stringent QPoS demands set. Rooted in illumination infrastructure reuse, VLP is destined to bring forth a solution for centimeter to decimetre-level localization at a limited cost. This manuscript investigated to which extent current VLP systems achieve these KPIs. Herein, it surveyed the VLP solution landscape, challenges, and accuracy error sources. It discusses VLP's adoption potential in different IPS application types, followed by a detailed overview of the different systems' associated QPoS indicators.

However, several challenges remain. Defining and standardizing a common set of QPoS metrics across IPS implementations is essential for fair benchmarking. This fairness can be further improved by establishing reference testbeds and measurement protocols that ensure all vendors run identical static-point and trajectory experiments, using standardized VLP transceiver modules. Hardware constraints such as limited LED modulation bandwidths and inappropriate driver circuits must be overcome to scale VLP beyond the lab. One promising solution is unmodulated VLP, which exploits light signals of opportunity. Rigorous 2.5-D and full 3-D localization evaluations are needed under realistic lighting conditions in realistic multipath environments (e.g., ceiling heights in industry $\gg$3 m). To meet the latency and reliability requirements of future industrial IoT applications, VLP will need further research into stable high-bandwidth LED sources and careful lighting planning that foresees redundant LoS links. Hybrid approaches, complementing VLP with IMU or 5G technology, might equally be able to meet these requirements. Incorporating reconfigurable intelligent surface (RIS)-based approaches could aid in increasing robustness and accuracy of VLP deployments.

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